Course Description:

This course provides an introduction to financial and management accounting principles and techniques. It covers financial statement preparation, analysis, and interpretation, as well as management accounting concepts, including cost behavior, budgeting, and performance management.

Course Objectives:

- Understand the role of accounting in business decision-making

- Prepare and analyze financial statements

- Apply management accounting techniques to support business decisions

- Understand cost behavior and cost-volume-profit analysis

- Prepare budgets and evaluate performance

CHAPTER ONE

INTRODUCTION TO ACCOUNTING CONCEPTS

The Accounting Concepts and Accounting Conventions have been developed over the years from experience, reason, usage and necessity and are generally accepted for accounting of transactions and preparation of Financial Statements. Accounting Concepts are the necessary assumptions, conditions or postulates upon which the accounting is based. They are developed to facilitate communication of the accounting and financial information to all the readers of the Financial Statements, so that all readers interpret the statements in the same meaning and context.

The Accounting Concepts are as follows:

1. **Entity Concept:**

For accounting purposes, an "organization" is treated as a separate entity from the "owners" or "stakeholders". This concept helps in keeping private affairs of the owners and stakeholders separate from the business affairs. **For example**, a ULB (urban local body or local govt entity) is a separate, independent and autonomous entity and is governed by a separate legislation and the regulations formed by it. The various stakeholders of the ULBs, including citizens, State Government, environmentalists, etc., do not own the ULBs. Thus, a separate Balance Sheet and Income & Expenditure Statement is prepared in respect of the operations of the ULB. This concept is applicable to all forms of organizations.

2. **Dual Aspect or Accounting Equivalence Concept :**

This concept follows from the Entity Concept. All entities own certain assets. Such assets are acquired through contributions of those who have provided the funds for the purpose. Funds are made available either through the surpluses of the entity or loans or payables. In a sense, such providers of funds are claimants to the assets. At any point in time, the assets will be equal to the claims. Since the claims on the assets could be those of "outsiders" (i.e. liabilities) or "owners" (i.e. capital, reserves, etc.), it results in the accounting equation: Assets = Own Funds + Liabilities.

3. Going Concern Concept

It is assumed that the organization will continue for a long time, unless and until it has entered into a state of liquidation. It is as per this concept, that the accountant does not take into consideration the market value of the assets while valuing them, irrespective of whether the market value is higher or lower than the book value. Similarly, depreciation on fixed assets is provided on the basis of expected lives of the assets rather than on their market values. Also, the financial statements are prepared at defined period-end to measure the performance of the entity during that period and not only on the closure or liquidation of the entity.

4. Money Measurement Concept

In accounting, every transaction is recorded in terms of money. Events or transactions that cannot be expressed in terms of money are not recorded in the books of accounts. Receipt of income, payment of expenses, purchase and sale of assets, etc., are monetary transactions that are recorded in the books of accounts. **For example**, the event of a machinery breakdown is not recorded as it does not have a monetary value. However, the expenditure incurred for the repair of the machinery can be measured in monetary value and hence is recorded.

5. Cost Concept

As per this concept, an asset is ordinarily recorded at the price paid to acquire it, i.e., at its cost and this cost is the basis for all subsequent accounting for the asset. The cost concept does not mean that the asset will always be shown at cost. This basically signifies that each time the financial statements are prepared, the fixed assets need not be revised and recorded at its realizable or replacement or market value. The assets recorded at cost at the time of purchase may systematically be reduced through depreciation.

6. Accounting Period Concept

An accounting period is the interval of time at the end of which the financial statements are prepared to ascertain the financial performance of the organization. Although the "going concern" concept stresses the continuing nature of the entity, it is necessary for an organization **(e.g. ULB)** to review how it is performing. The preparation of financial statements at periodic intervals helps in taking timely corrective action and developing appropriate strategies. The accounting period is normally considered to be of twelve months.

7. Accrual Concept

Under the cash system of accounting, the revenues and expenses are recorded only if they are actually received or paid in cash, irrespective of the accounting period to which they belong. But under the accrual concept, occurrence of claims and obligations in respect of incomes or expenditures, assets or liabilities based on happening of any event, passage of time, rendering of services, fulfillment (partially or fully) of contracts, diminution in values, etc., are recorded even though actual receipts or payments of money may not have taken place. In respect of an accounting period, the outstanding expenses and the prepaid expenses and similarly the income receivable and the income received in advance are shown separately in the books of accounts under the accrual method.

8. Periodic Matching of Cost and Revenue Concept

To ascertain the surplus or deficit made by the entity during an accounting period, it is necessary that the costs incurred are matched with the revenue earned by the entity during that accounting period. The matching concept is a corollary drawn from the accrual concept. To ascertain the correct surplus or deficit, it is necessary to make adjustments for all outstanding expenses, prepaid expenses, income receivable and income received in advance to correctly depict and match the income and expenditure relating to that accounting period.

9. Realization Concept

According to this concept, revenue should be accounted for only when it is actually realized or it has become certain that the revenue will be realized. This signifies that revenue should be recognized only when the services are rendered or the sale is affected. However, in order to recognize revenue, actual receipt of cash is not necessary. **What is important i**s that the organization should be legally entitled to receive the amount for the services rendered or the sale effected.

<u>Accounting Principles (THE DOUBLE ENTRY BOOKKEEPING)</u>

The double entry system is believed to have been formulated by an Italian Monk Luca Pacioli in the 1894. The system is based on the fact that every transaction has a double aspect or has two parties involved whereby one party gives while the other receives. That is every business transaction must be recorded in at least two accounts. One account is debited while the other is credited.

Therefore, the double entry principle states that, for every debit entry there must be a corresponding credit entry of the same amount.

APPLICATION OF DOUBLE ENTRY PRINCIPLE IN RECORDING TRANSACTION

Debit for increase **Credit for decrease**	**Debit for decrease** **Credit for increase**
ALL ➢ ASSETS, ➢ EXPENSES and ➢ DRAWINGS	**ALL** ➢ CAPITAL, ➢ LIABILITIES and ➢ REVENUE

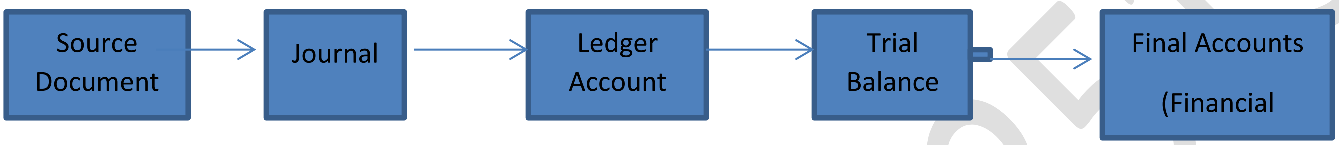
<u>FOLLOWING THE CLASSICAL SYSTEM</u>

Here we are going to see a system of recording where a single journal is used for all the prime recording. The entries in the journal are later posted into the ledger accounts. The balance of each account is calculated and a trial balance is extracted

The recording process of the classical system can be summarized as:

A. Information is picked from the source documents (such as invoice, cheques, receipts etc) and its recorded in the **journal**
B. The posting is then done into the **ledger accounts.**
C. From there, the **balanced broughtdown** in the ledger accounts are **extracted** and
D. These balanced brought downs are used to prepare the **trial balanced**,
E. from which the **final accounts** (that is profit and loss account, balanced sheet etc) are prepare(will be dealt with in the next chapter)

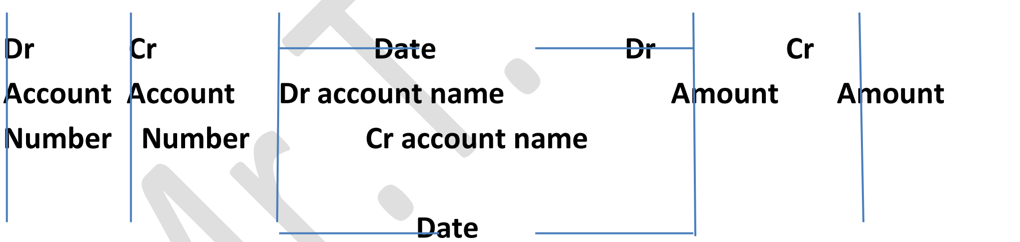
As said above in the general objective, the classical journal is illustrated below diagrammatically



<u>FORMAT OF THE CLASSICAL SYSTEM</u>

<u>1. CLASSICAL JOURNAL</u>

<u>Journal entries</u> are the first step in the accounting cycle and are used to record all **<u>business transactions</u>** and events in the accounting system. As business events occur throughout the accounting period, journal entries are recorded in the general journal to show how the event changed in the accounting equation. For example, when the company spends cash to purchase a new vehicle, the cash account is decreased or credited and the vehicle account is increased or debited.



<u>Example</u>

An enterprise has provided you with the following transactions for the year 2008.

Feb 2 purchased goods on credit 800,000 frs

Mar 3 sold goods by cheque 900,000 frs

May 5 paid transport by cash 50,000 frs

June 6 sold goods on credit 200,000 frs

<u>Required</u>

a)Record the transactions in a journal given that the enterprise uses the

SOLUTION

a) i. **Journal entry with the periodic inventory** **FCFA** **FCFA**

		2/2/2008		
601		Purchases	800,000	
	401	Account payables		800,000
		(To record credit purchases)		
		3/3/2008		
521		Bank	900,000	
	701	Sales		900,000
		(to record sales of good by cheque)		
		5/5/2008		
61		Transport	500,000	
	571	Cash		500,000
		(To record transport expense paid by cash)		
		6/6/2008		
411		Account receivables	200,000	
	701	Sales		200,000
		(To record credit sales of goods)		

EXAMPLE 1

You have been confronted with the following transactions for the month of January 2009:

Jan 1 Purchased goods by cash 500,000 frs

Jan 2 Sold goods by cash 400,000 frs

Jan 3 paid salaries by cash 300,000 FRS

Jan 4 Paid insurance by cash 200,000 frs

Required;

A. Record the transactions in a classical journal (ignore VAT)

B. Record the transactions in a classical journal (VAT rate 19.25%)

SOLUTION

A. Ignoring VAT

		1/1/2009		
601		Purchases of goods	500,000	
	571	Cash		500,000
		(to record cash purchases of goods)		
		2/1/2009		
571		Cash	400,000	
	701	Sales of goods		400,000
		(to record cash sales of goods)		
		3/1/2009		
661		salaries (direct remuneration)	300,000	
	571	Cash		300,000
		(to record payment of salaries by cash)		
		4/1/2009		
625		Insurance	200,000	
	571	Cash		200,000
		(To record payment of insurance by cash)		

B. VAT rate 19.25%

Calculation for VAT

Jan 1 VAT recoverable on Purchases = 19.25% x 500,000 = 96,250 frs

Jan 2 VAT invoice on sales = 19.25% x 400,000 = 77,000 frs

		1/1/2009		
601		Purchases of goods	500,000	
4452		VAT recoverable on Purchases	96,250	
	571	Cash		596,250
		(to record cash purchases of goods)		
		2/1/2009		
571		Cash	477,000	
	701	Sales of goods		400,000
	4431	VAT invoice on sales		77,000
		(to record cash sales of goods)		
		3/1/2009		
661		salaries (direct remuneration)	300,000	
	571	Cash		300,000
		(to record payment of salaries by cash)		
		4/1/2009		
625		Insurance	200,000	
	571	Cash		200,000
		(To record payment of insurance by cash)		

2. LEDGER ACCOUNT

Here, the OHADA account number is written beside the account name on the T-account. For instance, the T-account format of a cash account is represented as follows

571 Cash

Date	Description	Amount	Date	Description	Amount

EXAMPLE 2

You are provided with the following transactions for the month of February 2009

Feb 2 Bought goods on credit from Mona 500,000 frs

Feb 3 Sold goods by cash 400,000 frs

Feb 5 Purchased goods by cash 100,000 frs

Feb 7 Paid Mona by cash 200,000frs

Feb 11 Sales settled by cheque 150,000 frs

Feb 13 Paid electricity by cheque 50,000 frs

Feb 17 Credit sales to Lexy 175,000frs

Feb 19 Received a cheque from Lexy 125,000 frs

Feb 23 Paid mona by cheque 60,000 frs

Feb 27 paid salaries by cheque 80,000 Frs

Required

Show the classical ledger account using the T-format (ignore VAT)

3. TRIAL BALANCE

Be it a two, or four, or six or eight column trial balance under the classical system, the only column added will be the OHADA accounting number column.

Format Of A 2-Column Trial Balance

Name of the enterprise

Trial balance as at …./…/…..

Bal b/d

Account Number	**Description**	**Dr**	**Cr**
	Total		

Format Of A 4-Column Trial Balance

Name of the enterprise

Trial balance as at …./…/…..

Account Number	**Description**	**Movement**	**Balance b/d**

		Dr	Cr	Dr	Cr
	Total				

NB

Arrange the accounts number in ascending order that is from class 1 to class 9

Example 3

Prepare a four column classical trial balance for example 2 above

Example 4

MILI MINI enterprise has given you the following balances as at 1/1/2008

	FCFA
Stock of goods	1000,000
Cash	500,000
Bank	2,500,000
Suppliers	2,200,000
Customers	1,500,000
Personal Capital	8,000,000
Sales of goods	3,200,000
Purchases of goods	2,000,000
Buildings	1,900,000
Land	1,100,000
Transport equipment	2,800,000
Transport expenses	100,000

Required

Prepare a 2-column trial balance for MINI enterprise as at 1/1/2008

Solution

MILI MINI enterprise
Trial balance as at 1/1/2008

Account number	Details	Dr (FCFA)	Cr (FCFA)
103	Capital		8,000,000
22	Land	1,100,000	
23	Buildings	1,900,000	
24	Transport equipment	2,800,000	
311	Stock of goods	1,000,000	
401	Suppliers		2,200,000
411	Customers	1,500,000	
521	Bank	2,500,000	
571	Cash	500,000	
601	Purchases of goods	2,000,000	
61	Transport expenses	100,000	
701	Sales of goods		3,200,000
	Total	13,400,000	13,400,000

ADVANTAGES AND DISADVANTAGES OF THE CLASSICAL SYSTEM

- The **main advantages** of using the classical system are:
  - ✓ The system of recording is simple since only a single journal and a general journal are used.
  - ✓ It is less costly

- And the **main disadvantages** are
  - ✓ The recording in the journal is heavy, since only person is involved.
  - ✓ The manipulations of many accounts may slow down work especially when done by one person.

ASSIGNMENT

FREE BOY enterprise made available the following transactions for the month of January 2019

Jan 1 Started business with personal capital 1,000,000 frs cash

Jan 2 Purchased goods by cash 700,000 frs

Jan 3 Sold goods by cash 800,000 frs

Jan 5 Bought an office equipment by cash 500,000 frs

Jan 7 Purchased goods on credit from Zuma 300,000 frs

Jan 11 Sold goods on credit to Aruna 400,000 frs

Jan 13Recieved cash from Aruna 200,000 frs
Jan 17 Paid Zuma by Cash 100,000 frs

Required

Record the transaction in the classical journal, post to the ledger and extract a 4- column trial balance as at 17/01/2009 (VAT rate is 19.25%).

CHAPTER TWO

FINANCIAL STATEMENT ANALYSIS

GENERAL OBJECTIVES

A financial statement is an official document of the firm, which explores the entire financial information of the firm. The main aim of the financial statement is to provide information and understand the financial aspects of the firm. Hence, preparation of the financial statement is important as much as the financial decisions.

SPECIFIC OBJECTIVES

By the end of this chapter, the students should be able to

INTRODUCTION

Financial statements generally consist of two important statements:

(i) The income statement or profit and loss account.
(ii) Balance sheet or the position statement.

Financial statement analysis is interpreted mainly to determine the financial and operational performance of the business concern. A number of methods or techniques are used to analyse the financial statement of the business concern. But in this section, ratio analysis will be used

RATIO ANALYSIS

Ratio analysis is a commonly used tool of financial statement analysis. Ratio is a mathematical relationship between one number to another number. Ratio is used as an index for evaluating the financial performance of the business concern. An accounting ratio shows the mathematical relationship between two figures, which have meaningful relation with each other. Ratio can be classified into various types. Classification from the point of view of financial management is as follows:

- Liquidity Ratio

- Activity Ratio
- Solvency Ratio
- Profitability Ratio

SUMMARY OF THE FINANCIAL RATIOS

1. **PROFITABILITY RATIO**

a. **Gross Profit OR gross margin Ratio =** $\frac{\text{Gross profit}}{\text{Net sales}}$ x 100

Where Gross profit = Net sales – Cost of goods sold

And Net sales = sales – sales return- discount - allowance

Gross profit ratio evaluates how much gross profit is generated from sales

b. **Net profit margin =**$\frac{\text{Net income}}{\text{Net sales}} \text{ x } 100 \quad \text{Or} \quad \frac{\text{Net Profit after Tax}}{\text{Net sales}}$ x 100

This is also known as **Profit margin Or Return on sales.** It measures the percentage of income derived from dollar sales generated. The higher the net profit margin, the better**.**

c. **Operating profit ratio =** $\frac{\text{Operating net profit}}{\text{sales}}$ x 100

d. **Return on Assets =** $\frac{\text{Net income}}{\text{Average Total Assets}}$ x 100

Or $\frac{\text{Net Profit after Tax}}{\text{Average Total Assets}}$ x 100

It is abbreviated as **ROA,** and it measures management's efficiency in using assets to generate income

e. **Return on Investment =** $\frac{\text{Net income}}{\text{Shareholder's Equity or Funds}}$ x 100

Or $= \frac{\text{Net Profit after Tax}}{\text{Net sales}}$ x 100

It is abbreviated as **ROI,** and It measures the percentage of income derived from every dollar of owner's equity

f. **Return On Capital Employed =**$\frac{\text{Net Profit Before Interets and Tax}}{\text{Capital Employed}}$ x 100

It is abbreviated as **ROCE. Capital Employed** is also **known as working capital.**

And it is calculated as**:**

Capital employed = Total Assets – Current liability

(Non-current assets + Current assets) – Current Liability Or

Capital Employed = Non- current liability + Shareholder's Equity.

2. LIQUIDITY RATIO

It is also called as **short-term ratio.** This ratio helps to understand the liquidity in a business which is the potential ability to meet current obligations

a. **Current ratio or working capital ratio =** $\frac{\text{Current Assets}}{\text{Current liabilities}}$

It measures the ability of a company to pay short term obligations using current assets (cash, marketable securities, receivable, prepayment and inventory). The significant ratio is 2:1

b. **Acid test ratio or Quick ratio =**$\frac{\text{Current Assets} - \text{Inventory}}{\text{Current liabilities}}$

It measures the ability of a company to pay short term obligations using the more liquid types of current assets (cash, marketable securities, receivable, and prepayment).
The significant ratio is 1:1

c. **Cash ratio =** $\frac{(\text{Cash}+\text{Marketable securities})}{\text{Current liabilities}}$

It measures the ability of a company to pay it current liabilities using cash and marketable securities.

d. **Net working capital =** Current assets – current liabilities

It determines if a company can meet current obligations with it current assets and how much excess or deficiency there is.

3. LEVERAGE RATIO OR SOLVENCY RATIO

It measures the long-term obligation of the business concern. This ratio helps to understand, how the long-term funds are used in the business concern.

a. **Debt ratio =**$\frac{\text{Total liabilities}}{\text{Total Assets}}$

Measures the portion of company's assets that is financed by debt

b. **Equity ratio or Proprietor ratio** =$\frac{\text{Total Equity}}{\text{Total Assets}}$ Or $= \frac{\text{Shareholders Fund}}{\text{Total Assets}}$

Determine the portion of total assets provided by equity (i.e owner's contribution
And the company's accumulated profit)

Equity ratio can also be computed using the formula: 1- $\frac{\text{Total liabilities}}{\text{Total Assets}}$

The reciprocal of equity ratio is known as the equity multiplier, which is equals to the total assets divided by the total equity.

c. **Debt-Equity ratio**= $\frac{\text{Total liabilities}}{\text{Total Equity}}$ Or $\frac{\text{External Equity}}{\text{Internal Equity}}$

This evaluates the capital structure of the company. If it is more than 1 then it implies that the company is a leverage firm, but if less than 1 it is a conservative one

d. **Interest coverage ratio or Time interest earned**

$$= \frac{\text{EBIT}}{\text{Fixed Interest Charge}}$$

Where EBIT stands for Earnings before interest and tax measures the number of times interest expense is converted to income, and if the company can pay it interest expense using the profit generate.

e. **Average Credit Period Allowed (Debtor ratio)**

$=\frac{\text{Debtors}}{\text{Sales}}$x 360days or 52 weeks 0r 12 months

f. **Average credit period taken(Creditor Ratio)**

$= \frac{\text{Creditors}}{\text{Purchases}}$x 360days or 52 weeks 0r 12 months.

4. EFFICIENCY RATIO OR ACTIVITY RATIO

This ratio is helpful to understand the performance of the business concern

a. Inventory or Stock Turnover Ratio = $\frac{\text{Costof goods sold}}{\text{Average Inventory}}$

Where **Average inventory =** $\frac{\text{Opening inventory+Closing inventory}}{2}$

This represents the number of times inventory is sold and replaced. Take note that some authors use sales in the place of cost of goods sold in the above formula. A high ratio indicates that the company is efficient in managing it inventory

b. Receivables or Debtors Turnover Ratio=$\frac{\text{Net credit sales}}{\text{Average Account receivables}}$

This measures the efficiency of extending credit and collecting the same. It indicates the average number of times in a year a company collects it open accounts. A high rate implies efficiency credit and collection process.

c. Receivable turnover in days = $\frac{\text{360 days}}{\text{Recievable turnover ratio}}$

Also known as day's sales outstanding, or collection period. It measures the average number of days it takes a company to collect a receivable or it debt. The shorter the period, the better. Take note that some use 365 instead of 360 commercial days

d. Account payable turnover = $\frac{\text{Net credit purchase}}{\text{Average account payable}}$

Represents the number of times a company pays it account payable during a period. A low ratio is favored because it is better to delay payment as much as possible so that the money can be use for more productive purposes.

e. **Account payable turnover days** $= \frac{360}{\text{Account payable turnover}}$

Also known as day's payable outstanding, or payment period. It measures the average number of days spent before paying obligations to suppliers. The longer the period, the better.

f. **Working capital turnover or total assets turnover**$= \frac{\text{Net sales}}{\text{Net working capital or Average total assets}}$

Measures overall efficiency of a company in generating sales using tits assets. The formula is similar to ROA, except that net sales is used instead of capital

g. **Operating cycle =** Inventory turnover in days + Receivable turnover in days

Measures the number of days a company makes one complete cycle i.e purchase mechanize, sell the and later collect the amount due. A shorter operation cycle means that the company generates sales and collect the cash faster.

h. **Cash conversion cycle**= operating cycle – Account payable outstanding in days

This measures how fast company converts cash into more cash.it represents the number of days a company pays for purchases, sells them and then collect amount due. General like operating cycle, the shorter the cash conversion cycle the better.

Exercise

From the following balance sheet of Mr. Arvind Industries Ltd., as 31st March 2007.

Liabilities	frs.	Assets	frs.
Equity Share Capital	10,000	Fixed assets (less	26,000
7% Preference Share Capital	2,000	depreciation Rs. 10,000)	
Reserves and Surplus	8,000	Current Assets:	
6% Mortgage Debentures	14,000	Cash	1,000
Current Liabilities:		Investments (10%)	3,000
Creditors	1,200	Sundry debtors	4,000
Bills payable	2,000	Stock	6,000
Outstanding expenses	200		
Tax Provision	2,600		
	40,000		**40,000**

Other information:

1. Net sales frs. 60,000

2. Cost of goods sold frs. 51,600
3. Net income before tax frs. 4,000
4. Net income after tax frs. 2,000

Calculate appropriate ratios.

Solution

Short-term solvency ratios

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liability}} = \frac{14,000}{6,000} = 2.33:1$$

$$\text{Liquid Ratio} = \frac{\text{LiquidRatio}}{\text{Current Liability}} = \frac{8,000}{6,000} = 1.33:1$$

Long-term solvency ratios

$$\text{Proprietary ratio} = \frac{\text{Proprietor's funds}}{\text{Total Assets}} = \frac{20,000}{40,000} = 0.5:1$$

Proprietor's fund or Shareholder's fund=Equity share capital+Preference share capital+Reserve and surplus

= 10,000+2,000+8,000=20,000

$$\text{Debt-Equity ratio} = \frac{\text{External equities}}{\text{Internal equities}} = \frac{20,000}{20,000} = 1:1$$

$$\text{Interest coverage ratio} = \frac{\text{EBIT}}{\text{Fixed interest charges}} = \frac{4,000+840}{840} = 5.7\text{times}$$

Fixed interest charges = 6% on debentures of frs.14,000

= frs. 840

Activity Ratio

$$\text{Stock Turnover Ratio} = \frac{\text{Cost of Sales}}{\text{AverageInventory}} = \frac{51{,}600}{6{,}000} = 8.6\text{times}$$

As there is no opening stock, closing stock is taken as average stock.

$$\text{Debtors Turnover Ratio} = \frac{\text{Credit Sales}}{\text{Average Debtors}} = \frac{60{,}000}{6{,}000} = 10\text{times}$$

In the absence of credit sales and opening debtors, total sales is considered as credit sales and closing debtors as average debtors.

$$\text{Creditors turnover ratio} = \frac{\text{Credit Purchases}}{\text{AverageCreditors}} = \frac{43{,}200}{1{,}200} = 36\text{times}$$

In absence of purchases, cost of goods sold – gross profit treated as credit purchases and in the absence of opening creditors, closing creditors are treated as average creditors.

$$\text{Working Capital Turnover Ratio} = \frac{\text{Sales}}{\text{Net WorkingCapital}} = \frac{60{,}000}{8{,}000} = 7.5\text{times}$$

Profitability Ratios

$$\text{Gross profit ratio} = \frac{\text{GrossProfit}}{\text{Sales}} \times 100 = \frac{8{,}400}{60{,}000} \times 100 = 14\%$$

$$\text{Net profit ratio} = \frac{\text{Net Profit}}{\text{Sales}} \times 100 = \frac{2{,}000}{60{,}000} \times 100 = 3.33\%$$

In the absence of non-operating income, operating profit ratio is equal to net profit ratio.

Return of Investment = $\frac{\text{Net Profit afterTax}}{\text{Shareholder'sFund}} \times 100 = \frac{2,000}{20,000} \times 100 = 10\%$

Trend Analysis and Comparative Statements

Objective: To understand how to evaluate a company's financial performance over multiple periods to identify patterns in growth, profitability, and stability.

1. Comparative Statements (Horizontal Analysis)

- **Definition:** Comparative statements report financial figures for multiple period's side-by-side. It compares the same line items (e.g., Revenue, Cash, and Net Income) from one period to the next, usually shown in both absolute (dollar) amounts and percentages.
- **Purpose:**
  - **Identify Trends:** Detects the direction, speed, and extent of change over time.
  - **Variance Analysis:** Evaluates the effectiveness of budgeting by comparing actual performance to previous periods.
  - **Identify Red Flags:** For instance, if Accounts Receivable increases by 20% while Sales only grow by 5%, this warrants investigation into potential collection issues.
- **Key Components:**
- **Absolute Amount:** The dollar difference between periods.
- **Percentage Change:** The relative change, computed as:

$$\frac{\text{Current Year} - \text{Base Year}}{\text{Base Year}} \times 100$$

- **Limitations:** If a base year amount is zero or negative, a percentage change cannot be computed or is not meaningful.

2. Trend Analysis (Index-Number Trend Analysis)

- **Definition:** A form of horizontal analysis that examines financial data over a longer period (usually 3–5+ years) to identify long-term patterns.
- **Methodology:**
- Select a "Base Year" (usually the oldest year).
- Assign the base year a value of 100%.
- Divide subsequent years by the base year to determine trend percentages.

- **Uses:** Used to determine if a company's performance is consistently increasing, decreasing, or plateauing.
- **Example:** If 2021 Sales are (Base, 100%) and 2025 Sales are , the 2025 trend percentage is 150%, indicating a 50% growth over the period.

Common-Size Analysis (Vertical Analysis)

- **Definition:** While comparative statements look at changes *across time* (horizontal), Common-Size Analysis looks at the *composition* of a statement at a single point in time (vertical).
- **Methodology:**
  - **Income Statement:** All items are expressed as a percentage of **Net Sales**.
  - **Balance Sheet:** All items are expressed as a percentage of **Total Assets**.
- **Purpose:** Facilitates comparison between companies of different sizes or different industries.

➢ **limitations of Financial Statement Analysis**

Objective: To understand that financial statements are historical, subjective, and often incomplete, requiring analysts to use caution.

1. Historical Data (Backward Looking)

Financial statements are a "snapshot" of the past. They provide information on what *has happened*, which does not necessarily predict future performance.

- *Example:* A company may have had stellar growth, but a new competitor may make that growth irrelevant next year.

2. Inflation and Price Changes

Financial statements are prepared based on historical costs. They often do not adjust for inflation, meaning the true value of assets or the real value of profits can be distorted.

3. Varying Accounting Policies

Comparability is difficult because different companies use different, yet compliant, accounting methods.

- *Inventory:* FIFO vs. LIFO (FIFO tends to show higher income during inflation).
- *Depreciation:* Straight-line vs. Written-down value method.

4. Qualitative Factors Ignored

Financial statements are purely quantitative. They ignore critical qualitative factors that drive success, such as:

Quality of Management, Reputation and Brand Value, Employee Satisfaction and Morale. And Customer Loyalty.

5. "Window Dressing" and Manipulation

Companies may manipulate their financial statements to make them look better than they are (e.g., delaying expenses or accelerating revenue recognition).

6. No Real-Time Insights

Statements are often published months after the closing date, making the information outdated for rapid decision-making.

Summary of Mitigation Techniques

To overcome these limitations, analysts should:

Normalize data (remove one-time events), Use multiple years of data, not just two, Compare with industry averages, not just with the previous year, and Analyze cash flows in addition to net income.

CHAPTER THREE

FINANCIAL STATEMENT PREPARATIONS AND RESULTS

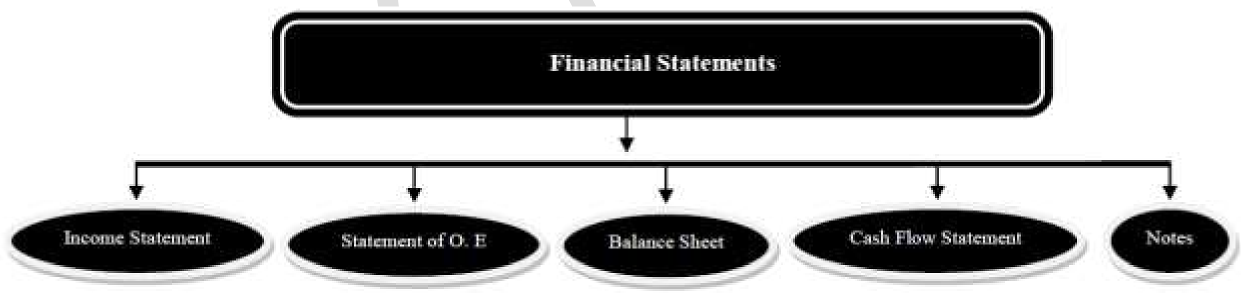
Financial statements are reports prepared and issued by company management to give investors and creditors additional information about a company's performance and financial standings. These reports are prepared in this order and are issued to the public as a full set of statements. This means they are not only published together, but they are also designed and intended to be read and used together. Since each statement only gives information about specific aspects of a company's financial position, it is important that these reports are used together.

What are Financial Statements?

Financial statements are reports prepared and issued by company management to give investors and creditors additional information about a company's performance and financial standings.
The five general purpose financial statements include:

- Income Statement
- Balance Sheet
- Statement of Stockholders Equity or Statement of Owner's Equity
- Statement of Cash Flows
- Notes to the Financial Statement

Here are some of the financial statements prepared by companies:



1. INCOME STATEMENT AND RESULTS

The income statement, also called the profit and loss statement, is a report that shows the income, expenses, and resulting profits or losses of a company during a specific time period. The income statement is the first financial statement typically prepared during the accounting cycle because the net income or loss must be calculated and carried over to the statement of owner's equity before other financial statements can be prepared.
The income statement calculates the net income of a company by subtracting total expenses from total income. This calculation shows investors and creditors the overall

profitability of the company as well as how efficiently the company is at generating profits from total revenues. The P&L is always for a specific period of time, such as a month, a quarter or a year. The periodic nature of the income statement is essential as this allows users to compare results for the company over similar periods of time, and to the results of other firms for the same period. Depending on the industry, year over year comparisons that eliminate seasonal variables can be especially useful.

FORMAT OF INCOME STATEMENT

Name of Company
Statement of income for the period ended 31/12/202X

	FCFA	FCFA	
Sales revenue or Turnover		x	
Sales returned or return outwards		(x)	
NET SALES			**XX**
LESS: Cost of Goods Sold			
Opening inventory		x	
Purchases		x	
Purchase returned or returned outwards		(x)	
Carriage inwards		x	
		x x	
Closing inventory		(x)	(xx)
GROSS PROFITS			**X X**
ADD: Other revenues			
Rent income		x	
Dividend income		x	
Discount received		x	
Interest received		x	
Commission received etc		x	**XX**
LESS: Other Expenses			
Carriage outwards		x	
Salaries and wages		x	
Rents and rates		x	
Insurance		x	
Depreciation		x	

Discount allowed	x	
Bad debts	x	
Stationery	x	
Interest payable	x	
Etc	x (XX)	
PROFIT BEFORE TAX		**XX**
Tax (x)		
NET PROFIT/ (LOSS) AFTER TAX		**XX/(XX)**

1. STATEMENT OF FINANCIAL POSITION OR BALANCE SHEET AND RESULTS

The balance sheet, also called the statement of financial position, is the third general purpose financial statement prepared during the accounting cycle. It reports a company's assets, liabilities, and equity at a single moment in time. You can think of it like a snapshot of what the business looked like on that day in time. Unlike the income statement, the balance sheet does not report activities over a period of time. The balance sheet is essentially a picture a company's recourses, debts, and ownership on a given day. This is why the balance sheet is sometimes considered less reliable or less telling of a company's current financial performance than a profit and loss statement. Annual income statements look at performance over the course of 12 months, whereas, the statement of financial position only focuses on the financial position of one day.

The balance sheet is basically a report version of the accounting equation also called the balance sheet equation where assets always equation liabilities plus shareholder's equity.

In this way, the balance sheet shows how the resources controlled by the business (assets) are financed by debt (liabilities) or shareholder investments (equity). Investors and creditors generally look at the statement of financial position for insight as to how efficiently a company can use its resources and how effectively it can finance them.

THE ACCOUNTING EQUATION (BALANCE SHEET EQUATION)

The three balance sheets elements cane be related in the following equation

ASSETS = CAPITAL + LIABILITY **or** *Assets = Owners Equity + Liabilities*

From the above

Capital or owner's equity = Assets – Liability and

Liability = Assets - Capital

EXAMPLES

1)A man starts a business with a capital of 2,000,000 frs and a total liability of 1,500,000 frs. Determine the value of the assets of that business.

Solution
Capital = 2,000,000   liability = 1,500,000 frs
And A = C +L = 2000000+1500000= 3500000 FRS

2) The assets of a business is worth 800,000frs . A liability of 300,000 frs has been engaged. What is the capital of the business?
Solution
Assets = 800,000 frs   liability = 300,000 frs
Capital = assets – liability = 800,000 – 300,000= 500,000 frs

3) Determine the liability of an enterprsie that possesses a capital of 1,200,000 frs and a total assets of 1,900,000 frs.
Solution
Capital = 1,200,000 frs assets = 1,900,000frs
Liability = assets- capital = 1,900,000 – 1,200,000 = 700,000 frs.

<u>FORMAT OF A STATEMENT OF FINANCIAL POSITION OR BALANCE SHEET</u>

Name of company
Balance sheet as at 31st December 201X

	COST	ACCUMULATED DEPRECIATION	NBV
	FCFA	FCFA	FCFA
<u>FIXED ASSETS OR NON CURRENT ASSETS</u>:			
INTANGIBLE ASSETS			
Goodwill	x	(x)	x
Patent	x	(x)	x
TANGIBLE ASSETS			

Land and building	x	(x)	x
Plant and Machinery	x	(x)	x
Equipment	x	(x)	x
Fixtures and Fittings	x	(x)	x
Motor Van etc	x	(x)	x
TOTAL FIXED ASSETS			**XX**
CURRENT ASSETS:			
Inventory or stock of goods	x		
Receivables or debtors	x		
Cash at bank	x		
Cash at hand	x		
Marketable securities	x		
Prepaid expenses or prepayments etc	x		
TOTAL CURRENT ASSEST			**XX**
TOTAL ASSETS			**XXX**
LESS: CURRENT LIABILITIES:			
Bank overdraft	x		
Short term Creditors	x		
Accruals	x		
Short term loans	x		
Accounts Payables etc	x **(XX)**		
WORKING CAPITAL		**XXXX**	
FINANCED BY:			
OWNER'S EQUITY			
Capital	x		
Retained Earnings	x		
Net Profit	x		
Share Premium	x		
Drawings	(x) **XX**		
NON CURRENT LIABILITIES:			
Long term loan	x		
Notes payables	x		

Bonds payables	x	
Debentures	<u>(x)</u>	<u>**XX**</u>

<u>**XXXX**</u>

EXAMPLE 1

AFUACAM enterprise has provided you with the following balances as at 31st December 2009

	FCFA
Stock at 1/10/2009	900,000
Sales	4,900,000
Purchases	2,500,000
Return inwards	107,000
Return outwards	210,000
Carriage inwards	107,000
Carriage outwards	111,000
Discount allowed	75,000
Discount received	
Discount received	25,000
Rents and rates	390,000
Insurance	250,000
Telephone	100,000
Salaries	410,000
Transport	200,000

<u>**Required**</u>

Prepare the trading and profit and loss account for the year ended 31 of December 2009, given that the stock as at 31 of December is worth 120,000FCFA.

SOLUTION

AFUACAM enterprise

Trading profit and loss account for the year ended 31st December 2009

	FCFA	FCFA
Sales	4,900,000	
Return inwards	<u>(107,000)</u>	4,793,000
Less Cost of Goods Sold:		
Opening stock	900,000	

Purchases	2,500,000	
Return outwards	(210,000)	
Carriage inwards	107,000	
Cost of good available for sales	3,297,000	
Closing stock	(120,000)	(3,177,000)
Gross profit		**1,616,000**
Add other income:		
Discount received		25,000
Less expenses:		
Carriage outwards	111,000	
Discount allowed	75,000	
Rents and rates	390,000	
Insurance	250,000	
Telephone	100,000	
Salaries	410,000	
Transport	200,000	(1,536,000)
Net profit		**105,000**

ADJUSTMENTS FOR FINANCIAL STATEMENTS

There are several adjustments to be made at the end of the financial year before financial statements are prepared. These adjustments incudes **accruals prepayments, depreciation, bad debt, provision for doubtful debts, capital expenditures, revenue expenditures** amongst others

1) ACCRUALS

These are **expenses owing** by the end of the financial year. They are also called **outstanding expenses or expense creditors**. In **the profit and loss account**, accruals are added **as expenses** while in **the balance sheet**, they are treated as **current liabilities.**

2) PREPAYMENT OR PREPAID EXPENSES

These are expenses made **in advanced** by the end of the financial year. They are also called **expense debtors**. In **the profit and loss account**, prepayments are **subtracted** from **the expense** while in **the balance sheet**; they are treated as **current assets**.

3) DEPRECIATION

These are reductions made on the value of fixed assets at the end of the financial year.in the profit and loss account, depreciation are treated as expenses for that period, while in the balance sheet, it is accumulated (accumulated depreciation) , and then subtracted from the fixed assets.

Depreciation is commonly c**alculated** using either the **straight line method (also known as constant method)** . Below is a formula

Depreciation expense = Original value x Rate = O.V x R

Or

$$= \frac{\textbf{Original value}}{\textit{useful life}} \qquad = \frac{\textbf{O.V}}{n}$$

Or

$$= \frac{\textbf{Original value} - \textbf{scarp value}}{\textit{useful life}} \qquad = \frac{\textbf{O.V} - \textbf{s,v}}{n}$$

Scarp value is also known as residual value or salvage value.

Accumulated depreciation = depreciation expense X number of years

And therefore,

NET BOOK VALUE (NBV) = Original value – Accumulated depreciation

4) PROVISION

This is a **probable** loss in the value of an asset. In **the profit and loss account** provisions are treated as **expense** while in **the balance sheet**, they are **subtracted from the assets.**

5) BAD DEBTS

These are debts that cannot be recovered, either because the debtor is bankrupt, death or has disappeared without trace. In **the profit and loss account**, bad debts are treated as **expense**s while in **the balance sheet**; they are **subtracted from the debtors account.**

6) PROVISION FOR DOUBTFUL DEBTS

It can also be called **provision for bad debt**.it is created only when the debtor has become doubtful. That’s therefore a **debt may not be recovered**. It has the **same treatment as bad debt** both in the profit and loss account and balance sheet. That is **in the profit and loss account**, doubtful debts are treated **as expenses** while in the **balance sheet**; they are **subtracted from the debtors account.**

7) **CAPITAL EXPENDITURES**

These are expenditures **on fixed assets** that aim at **increasing the earning capacity** of that asset. These fixed assets are also known as capital items such as buildings, van, equipment, plants and machinery. Such items have long life span and can last in the enterprise for more than one year. This expenditure is treated in the **balance sheet** by **adding the additional value to the specific asset.**

8) **REVENUE EXPENDITURES**

These are expenditures on **current assets** that aim at **maintaining the earning capacity** of the asset **(E.g repairs and renewals**). It also relates to the **daily running of the business**. This expenditure is treated in the **profit and loss account as an expense**.

CHAPTER FOUR

MANAGEMENT ACCOUNTING

Explain the role of Management Accountant in the Managerial Functions?

Also known as corporate accountants, management accountants work within one specific company. The role of the management accountant is to perform a series of tasks to ensure their company's financial security, handling essentially all financial matters and thus helping to drive the business's overall management and strategy.

Management accountants are key figures in determining the status and success of a company. Some choose to become a Certified Management Accountant (CMA), a similar credential to CPA, but with a greater focus on cost accounting, financial planning, and management issues. Job responsibilities can range widely. Depending on the company, accountant's level of experience, the time of year and the type of industry, he could find himself doing any of the following tasks:

- Budgeting
- Handling taxes
- Managing assets to helping determine compensation and benefits packages
- Aiding in strategic planning

Role of Management Accountant:

Consistent with other roles in today's corporation, management accountants have a dual reporting relationship. As a strategic partner and provider of decision based financial and operational information, management accountants are responsible for managing the business team and at the same time having to report relationships and responsibilities to the corporation's finance organization.

The activities management accountants provide inclusive of forecasting and planning, performing variance analysis, reviewing and monitoring costs inherent in the business are ones that have dual accountability to both finance and the business team. **Examples of tasks** where accountability may be more meaningful to the business management team vs. the corporate finance department are the development of new product costing, operations research, business driver metrics, sales management score carding, and client profitability analysis. (See financial modeling.) Conversely, the preparation of certain financial reports, reconciliations of the financial data to source systems, risk and regulatory reporting will be more useful to the corporate finance team as they are charged with aggregating certain financial information from all segments of the corporation.

In corporations that derive much of their profits from the information economy, such as banks, publishing houses, telecommunications companies and defense contractors, IT costs are a significant source of uncontrollable spending, which in size is often the greatest corporate cost after total compensation costs and property related costs. A function of management accounting in such organizations is to work closely with the IT department to provide IT cost transparency.

Given the above, one widely held view of the progression of the accounting and finance career path is that financial accounting is a steppingstone to management accounting. Consistent with the notion of value creation, management accountants help drive the success of the business while strict financial accounting is more of a compliance and historical endeavor.

The role of the management accountant is to perform a series of tasks to ensure their company's financial security, handling essentially all financial matters and thus helping to drive the business's overall management and strategy.

Management accountants are key figures in determining the status and success of a company. Management accountants give greater focus on cost accounting, financial planning, and management issues. In **Management accounting *or* managerial accounting,** managers use the provisions of accounting information in order to better inform themselves before they decide matters within their organizations, which allows them to better manage and perform control functions.

If you like keeping track of a company's income and expenses but also want to hold a position with significant responsibility and authority, management accounting could be the job for you. This lesson will teach you about the profession of management accounting, from a management accountant's job responsibilities, skill set and formal educational requirements to the professional designations that can help you get ahead, as well as the career ladder for a management accounting job.

Job Description

Management accountants work for public companies, private businesses and government agencies. They may also be called cost accountants, managerial accountants, industrial accountants, private accountants or corporate accountants, but all perform similar functions within a company. Indeed, preparing data for use within a company is one of the features that distinguishes a management accountant's job from other types of accounting jobs, such as public accounting.

Instead, you'll be recording and crunching numbers for internal review to help companies budget and perform better. In conjunction with other managers in the company, you may help the company choose and manage its investments. Management accountants are risk managers, budgeters, planners, strategists and decision makers. They do the work that helps the company's owner, manager or board of directors make decisions.

As a management accountant, you'll likely supervise lower-level accountants who handle a company's basic accounting tasks, such as recording income and expenses, tracking tax liabilities and using these data to prepare income statements, cash flow statements and balance sheets, but in a smaller firm, you might perform these tasks yourself. A management accountant will analyze these basic data and make forecasts, budgets, performance measurements and plans, then present them to senior management to assist in its operational decision making.

A management accountant may also identify trends and opportunities for improvement, analyze and manage risk, arrange the funding and financing of operations and monitor and enforce compliance. They might also create and maintain a company's financial system and supervise its bookkeepers and data processors. Further, management accountants may have an area of expertise, such as taxes or budgeting.

- **What are the advantages and limitations of Financial Accounting?**

Financial accounting is the field of accountancy concerned with the preparation of financial statements for decision makers such as stockholders, suppliers, banks, employees, government agencies, owners, and other stakeholders. The practice is used to prepare accounting information for people outside the organization or not involved in the day-to-day running of the company. It also helps people to store information on a company or organization that would be difficult to memorize.

It allows people to measure the profitability and value of a business, rather like having a dossier of the business's transactions you can see how well or how poorly the company is doing. It also helps when attempting to compile a company report for those outside the company, this might be for prospective buyers or investors, who wish to see a full and detailed account before becoming involved. Financial accounting and the information it generates can act like a barometer for an organization, allowing achievements and losses to be measured against rivals and competitors and produce a clear picture of where a firm stands in its particular field.

Advantages:

1. Accounting supplies numerical information to the institution relating to its management and administration
2. Exact results of the institution are disclosed through accounting
3. The firm can ascertain the financial status of the business operation
4. Firm can compare the financial position of two/more years
5. Books accounts are very valuable documents
6. Proper accounting makes the firm credible to other party
7. Tax authority can assess taxes for the firm using the accounting information
8. Firm can determine the actual assets and liabilities
9. Using accounting data a firm can formulate policy and take many decision on future operation

Limitations:

(i) No detail information about cost:

Financial Accounting provides information as a whole in terms of income, expenses, assets and liabilities. It does not provide detail of cost involved by departments, processes, products, services or other unit of activity within the organization.

(ii) No cost control method:

It does not have proper mechanism to control expenditure on various elements of cost, viz, material and labour. As a result, misappropriation, wastage and losses of materials are left unchecked. Proper utilization of labour becomes impossible and suitability of different labour incentive plans goes without evaluation.

(iii) No information on efficiency:

It does not have a system to judge the» efficiency in the use of material, labour and overhead costs of the organization in comparison to the standard fixed for their use.

(iv) No classification of expenses:

It does not classify expenses as direct and indirect or fixed and variable. Besides, these are also not allocated to different stages of production or departments or processes to show the controllable and uncontrollable items on overhead cost.

(v) Historical in nature:

It is prepared at the end of the accounting ♦period. It fails to provide day-to-day information to pre-determine cost.

(vi) Not helpful in price fixation:

It does not provide adequate cost information to fix up the price of products manufactured and service rendered by the organization.

(vii) No analysis of losses:

It does not provide detail information about the reasons of losses. It also does not help to determine the variations in the cost between different working times, idle time and seasonal conditions of the industry.

(viii) No technique to evaluate alternative methods:

In planning expansions contraction of plants, equipment, products and processes it is not poses to calculate and compare the profitability of alternatives with the help Financial Accounting.

(ix) No cause and effect analysis:

As financial account fails to provide o and profit information product-wise, the causes of profit or loss cannot effectively determined and analyzed.

(x) No data for comparison:

It does not provide data to facilitate compare of costs of operation of the firm with other firms in the industry.

Cost Accounting is developed from within the accounting process to overcoat the limitations of Financial Accounting and it helps in calculating, controlling and reducing cost.

- **TRIAL BALANCE**

What is Trial Balance? Explain its purpose and limitations and give format of the Trial Balance.

Trial Balance is a list of closing balances of ledger accounts on a certain date and is the first step towards the preparation of financial statements. It is usually prepared at the end of an accounting period to assist in the drafting of financial statements. Ledger balances are segregated into debit balances and credit balances. Asset and expense accounts appear on the debit side of the trial balance whereas liabilities, capital and income accounts appear on the credit side. If all accounting entries are recorded correctly and all the ledger balances are accurately extracted, the total of all debit balances appearing in the trial balance must equal to the sum of all credit balances.

Purpose of a Trial Balance

Trial Balance acts as the first step in the preparation of financial statements. It is a working paper that accountants use as a basis while preparing financial statements.

Trial balance ensures that for every debit entry recorded, a corresponding credit entry has been recorded in the books in accordance with the double entry concept of accounting. If the totals of the trial balance do not agree, the differences may be investigated and resolved before financial statements are prepared. Rectifying basic accounting errors can be a much lengthy task after the financial statements have been prepared because of the changes that would be required to correct the financial statements.

Trial balance ensures that the account balances are accurately extracted from accounting ledgers.

Trail balance assists in the identification and rectification of errors.

Example
Following is an example of what a simple Trial Balance looks like:

ABC LTD Trial Balance as at 31 December 2016		
Account Title	Debit	Credit
	frs. '000'	frs. '000'
Share Capital		15,000
Furniture & Fixture	5,000	
Building	10,000	
Creditor		5,000
Debtors	3,000	
Cash	2,000	
Sales		10,000
Cost of sales	8,000	
General and Administration Expense	2,000	
Total	30,000	30,000

Title provided at the top shows the name of the entity and accounting period end for which the trial balance has been prepared.

Account Title shows the name of the accounting ledgers from which the balances have been extracted.

Balances relating to assets and expenses are presented in the left column (debit side) whereas those relating to liabilities, income and equity are shown on the right column (credit side).

The sum of all debit and credit balances is shown at the bottom of their respective columns.

Limitations of a trial balance

Trial Balance only confirms that the total of all debit balances match the total of all credit balances. Trial balance totals may agree in spite of errors. An example would be an incorrect debit entry being offset by an equal credit entry. Likewise, a trial balance gives no proof that certain transactions have not been recorded at all because in such case, both debit and credit sides of a transaction would be omitted causing the trial balance totals to still agree. Types of accounting errors and their effect on trial balance are more fully discussed in the section on Suspense Accounts.

CHAPTER FIVE
COST ACCOUNTING

What is "Cost Accounting"? State its importance.

A method of accounting in which all costs incurred in carrying out an activity or accomplishing a purpose are collected, classified, and recorded. This data is then summarized and analyzed to arrive at a selling price, or to determine where savings are possible.
In contrast to financial accounting (which considers money as the measure of economic performance) cost accounting considers money as the economic factor of production.
Cost accounting is a process of collecting, analyzing, summarizing and evaluating various alternative courses of action. Its goal is to advise the management on the most appropriate course of action based on the cost efficiency and capability. Cost accounting provides the detailed cost information that management needs to control current operations and plan for the future.
Since managers are making decisions only for their own organization, there is no need for the information to be comparable to similar information from other organizations. Instead, information must be relevant for a particular environment. Cost accounting information is commonly used in financial accounting information, but its primary function is for use by managers to facilitate making decisions.

Unlike the accounting systems that help in the preparation of financial reports periodically, the cost accounting systems and reports are not subject to rules and standards like the Generally Accepted Accounting Principles. As a result, there is wide variety in the cost accounting systems of the different companies and sometimes even in different parts of the same company or organization.

SECTION 1

Importance of Cost Accounting

The limitation of financial accounting has made the management to realize the importance of cost accounting. The importance of cost accounting is as follows:

Importance to Management

Cost accounting provides invaluable help to management. It is difficult to indicate where the work of cost accountant ends and managerial control begins. The advantages are as follows:

i. **Helps in ascertainment of cost**
   Cost accounting helps the management in the ascertainment of cost of process, product, Job, contract, activity, etc., by using different techniques such as Job costing and Process costing.
ii. **Aids in Price fixation**
   By using demand and supply, activities of competitors, market condition to a great extent, also determine the price of product and cost to the producer does play an important role. The producer can take necessary help from his costing records.
iii. **Helps in Cost reduction**
   Cost can be reduced in the long-run when cost reduction programme and improved methods are tried to reduce costs.
iv. **Elimination of wastage**
   As it is possible to know the cost of product at every stage, it becomes possible to check the forms of waste, such as time and expenses etc., are in the use of machine equipment and material.
v. **Helps in identifying unprofitable activities**
   With the help of cost accounting the unprofitable activities are identified, so that the necessary correct action may be taken.
vi. **Helps in checking the accuracy of financial account**
   Cost accounting helps in checking the accuracy of financial account with the help of reconciliation of the profit as per financial accounts with the profit as per cost account.
vii. **Helps in fixing selling Prices**. It helps the management in fixing selling prices of product by providing detailed cost information.

Importance to Employees

Worker and employees have an interest in which they are employed. An efficient costing system benefits employees through incentives plan in their enterprise, etc. As a result both the productivity and earning capacity increases.

Cost accounting and creditors

Suppliers, investor's financial institution and other moneylenders have a stake in the success of the business concern and therefore are benefited by installation of an efficient costing system. They can base their judgment about the profitability and prospects of the enterprise upon the studies and reports submitted by the cost accountant.

Importance to National Economy

An efficient costing system benefits national economy by stepping up the government revenue by achieving higher production. The overall economic developments of a country take place due to efficiency of production.

Data Base for operating policy

Cost Accounting offers a thoroughly analyzed cost data which forms the basis of formulating policy regarding day to day business.

SECTION 2 Discuss various types of costs in detail?

In production, research, retail, and accounting, a cost is the value of money that has been used up to produce something, and hence is not available for use anymore. In business, the cost may be one of acquisition, in which case the amount of money expended to acquire it is counted as cost. In this case, money is the input that is gone in order to acquire the thing. This acquisition cost may be the sum of the cost of production as incurred by the original producer, and further costs of transaction as incurred by the acquirer over and above the price paid to the producer. Usually, the price also includes a mark-up for profit over the cost of production.

An amount that has to be paid or given up in order to get something is called as cost. In business, cost is usually a monetary valuation of (1) effort, (2) material, (3) resources, (4) time and utilities consumed, (5) risks incurred, and (6) opportunity forgone in production and delivery of a good or service. All expenses are costs, but not all costs (such as those incurred in acquisition of an income-generating asset) are expenses.

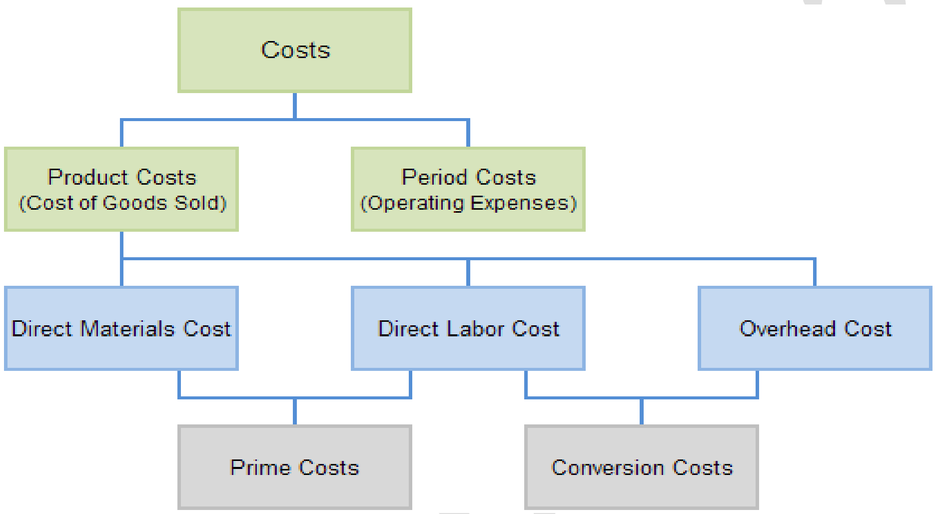
Definition of Cost

"Something of value, usually an amount of money, given up in exchange for something else, usually goods or services. All expenses are costs, but not all costs are expenses. (An expense is the cost of resources used to produce revenue.) As a verb, cost means to estimate the amount of money needed to produce a product or perform a service."

Cost and Cost Classifications

Cost is a sacrifice of resources to obtain a benefit or any other resource. For example in production of a car, we sacrifice material, electricity, the value of machine's life (depreciation), and labor wages etc. Thus these are our costs.

Costs are usually classified as follows:



1. **Product Costs Vs. Period Costs**

Product costs are costs assigned to the manufacture of products and recognized for financial reporting when sold. They include direct materials, direct labor, factory wages, factory depreciation, etc.

Period costs are on the other hand are all costs other than product costs. They include marketing costs and administrative costs, etc.

2. **Breakup of Product Costs**

The product costs are further classified into:

Direct materials: Represents the cost of the materials that can be identified directly with the product at reasonable cost. For example, cost of paper in newspaper printing, cost of

Direct labor: Represents the cost of the labor time spent on that product, for example cost of the time spent by a petroleum engineer on an oil rig, etc.

Manufacturing overhead: Represents all production costs except those for direct labor and direct materials, for example the cost of an accountant's time in an organization, depreciation on equipment, electricity, fuel, etc.

The product costs that can be specifically identified with each unit of a product are called direct product costs. Whereas those which cannot be traced to a specific unit are indirect product costs. Thus direct material cost and direct labor cost are direct product costs whereas manufacturing overhead cost is indirect product cost.

3. **Prime Costs Vs. Conversion Costs**

Prime costs are the sum of all direct costs such as direct materials, direct labor and any other direct costs.

Conversion costs are all costs incurred to convert the raw materials to finished products and they equal the sum of direct labor, other direct costs (other than materials) and manufacturing overheads.

4. **Fixed Costs Vs. Variable Costs**

Fixed costs are costs which remain constant within a certain level of output or sales. This certain limit where fixed costs remain constant regardless of the level of activity is called relevant range. For example, depreciation on fixed assets, etc.

Variable costs are costs which change with a change in the level of activity. Examples include direct materials, direct labor, etc.

5. **Sunk Costs Vs. Opportunity Costs**

The costs discussed so far are historical costs which means they have been incurred in past and cannot be avoided by our current decisions. Relevant in this regard is another cost classification, called sunk costs. Sunk costs are those costs that have been irreversibly incurred or committed; they may also be termed unrecoverable costs.

In contrast to sunk costs are opportunity costs which are costs of a potential benefit foregone. For example the opportunity cost of going on a picnic is the money that you would have earned in that time.

CHAPTER SIX
MANAGEMENT ACCOUNTING AND BUDGETING

Describe the nature and scope of Management Accounting?

Management accounting can be viewed as Management-oriented Accounting. Basically it is the study of managerial aspect of financial accounting, "accounting in relation to management function". It shows how the accounting function can be re-oriented so as to fit it within the framework of management activity. The primary task of management accounting is, therefore, to redesign the entire accounting system so that it may serve the operational needs of the firm.

If furnishes definite accounting information, past, present or future, which may be used as a basis for management action. The financial data are so devised and systematically development that they become a unique tool for management decision.
The term “Management Accounting”, observe, Broad and Carmichael, covers all those services by which the accounting department can assist the top management and other departments in the formation of policy, control of execution and appreciation of effectiveness. This definition points out that management is entrusted with the primary task of planning, execution and control of the operating activities of an enterprise. It constantly needs accounting information on which to base its decision. A decision based on data is usually correct and the risk of erring is minimized.

Definitions

“Any form of accounting which enable the business to be conducted more efficiently may regarded as Management Accounting”
“Accounting for Management”

According to the Institute of Management Accountants (IMA): "Management accounting is a profession that involves partnering in management decision making, devising planning and performance management systems, and providing expertise in financial reporting and control to assist management in the formulation and implementation of an organization's strategy"

The Management Accounting Practices Committee (MAPC) of the National Accounting Association (NAA) in the United States defines Management Accounting as “the process of identification, measurement, accumulation, analysis, preparation and communication of financial information used by management to plan, evaluate and control within an organization and to ensure appropriate use of accountability for its resources.”
Management Accounting can also be defined as “the application of professional skills in preparation and presentation of accounting information in such a way as to assist to management in the formulation of policies and in the planning and control of the operations of the undertaking.” (Sizer 1996).

Nature of Management Accounting

The term management accounting is composed of 'management' and 'accounting'. The word 'management' here does not signify only the top management but the entire personnel charged with the authority and responsibility of operating an enterprise. The task of management accounting involves furnishing accounting information to the management, which may base its decisions on it. It is through management accounting that the management gets the tools for an analysis of its administrative action and can lay suitable stress on the possible alternatives in terms of costs, prices and profits, etc. but it should be understood that the accounting information supplied to management is not the sole basis for managerial decisions. Along with the accounting information, management takes into consideration or weighs other factors concerning actual execution. For reaching a final decision, management has to apply its common sense, foresight, knowledge and experience of operating an enterprise, in addition to the information that is already has. The word 'accounting' used in this phrase should not lead us to believe that it is restricted to a mere record of business transactions i.e., book keeping only. It has indeed a 'macro-economic approach'. As it draws its raw material from several other disciplines like costing, statistics, mathematics, financial accounting, etc., it can be called an interdisciplinary subject, the scope of which is not clearly demarcated. Other fields of study, which can be covered by management accounting, are political science, sociology, psychology, management, economics, statistics, law, etc. A knowledge of political science helps to understand authority relationship and responsibility identification in an organization. A study of sociology helps to understand the behaviour of man in groups. Psychology enables us to know the mental make-up of employers and employees. A knowledge of these subjects helps to increase motivation, and to control the actions of the people who are ultimately responsible for costs. This builds a better employer-employee relationship and a sound morale. The subject of management reveals the processes involved in the art of managing, a knowledge of economics assists in the determination of optimum output in the forecasting of sales and production, etc., and also makes it possible to analyze management action in terms of cost revenues, profits, growth, etc. It is with the help of statistics that this information is presented to the management in a form that can be assimilated. The subject of management accounting also encompasses the subject of law, knowledge of which is necessary to find out if the management action is ultra-vires or not.

It is, therefore, a wide and diverse subject.

Management accounting has no set principles such as the double entry system of bookkeeping. In place of generally accepted accounting principles, the philosophy of cost benefit analysis is the core guide of this discipline. It says that no accounting system is good or bad but is can be considered desirable so long as it brings incremental benefits in excess of its incremental costs. Applying management accounting principles to financial matters can arrive at no single perfect solution. It is, therefore, an inexact science, which uses its own conventions rather than standardized principles. Since management accounting is managerially oriented, its data is selective in nature. It focuses on potential opportunities rather than opportunities lost. The data is operative in nature catering to the operational needs of a firm. It details events, monetary and non-monetary. The nature of data, the form of presentation and its duration are mainly determined by managerial needs. It is quite frequently reported as it is meant for internal uses and managerial control. An accountant should look at his enterprise from the management's point of view. Whenever he fails to do that he ceases to be a management accountant.

Scope of Management Accounting

The scope or field of management accounting is very wide and broad based and it includes a variety of aspects of business operations. The main aim of management accounting is to help management in its functions of planning, directing, controlling and areas of specialization included within the admit of management accounting. The scope of management accounting can be studied as follows:

1. Financial Accounting

Financial accounting forms the basis for analysis and interpretation for furnishing meaningful data to the management. The control aspect is based on financial data and performance evaluation, on recorded facts and figures. So, management accounting is closely related to financial accounting in many respects.

2. Cost Accounting

Cost accounting is the process and techniques of ascertaining cost. Planning, decision making and control are the basic managerial functions. The cost accounting system provides the necessary tool for carrying out such functions efficiently. The tools includes standard costing, inventory management, variable costing etc.

3. Budgeting and Forecasting

Budgeting means expressing the plans, policies and goals of the firm for a definite period in future. Forecasting on the other hand, is a prediction of what will happen as a result of a given set of circumstances. Forecasting is a judgment whereas the budgeting is an organizational object. These are useful for management accounting in planning.

4. Inventory Control

Inventory is necessary to control from the time it is acquire till its final disposal as it involves large sum. For controlling inventory, management should determine different level of stock. The inventory control technique will be helpful for taking managerial decisions.

5. Statistical Method

Statistical tools not only make the information more impressive, comprehensive and intelligible but also are highly useful for planning and forecasting.

6. Interpretation of Data

Analysis and interpretation of financial statements are important part of management accounting. After analyzing the financial statements, the interpretation is made and the reports drawn from this analysis are presented to the management. Interpreting the accounting data to the authorities in the management is the principal task of management accounting.

7. Reporting To Management

The interpreted information must be communicated to those who are interested in it. The report may cover Profit and Loss Account, Cash Flow and Funds Flow statements etc.

8. Internal Audit and Tax Accounting

Management accounting studies all the tax matters to assist the management in investment decisions vis-a-vis tax planning as a resource to enjoy tax relief.

Internal audit system is necessary to judge the performance of every department. Management is able to know deviations in performance through internal audit. It also helps management in fixing responsibility of different individuals.

9. Methods of Procedures

This includes maintenance of proper data processing and other office management services. It may have to deal with filing, copying, duplicating, communicating and management information system and also may have to report about the utility of different office machines.

Distinguish between Financial Accounting and Cost Accounting.

The Financial Accounting is the process of recording, summarizing and reporting the myriad of transactions from a business, so as to provide an accurate picture of its financial position and performance. The primary objective of financial accounting is the preparation of financial statements - including the balance sheet, income statement and cash flow statement - that summarizes the company's operating performance over a particular period, and financial position at a specific point in time. These statements - which are generally prepared quarterly and annually, and in accordance with Generally Accepted Accounting Principles (GAAP) - are aimed at external parties including investors, creditors, regulators and tax authorities.

Cost accounting is a process of collecting, analyzing, summarizing and evaluating various alternative courses of action. Its goal is to advise the management on the most appropriate course of action based on the cost efficiency and capability. Cost accounting provides the detailed cost information that management needs to control current operations and plan for the future. This data is then summarized and analyzed to arrive at a selling price, or to determine where savings are possible.

Even though, cost accounting and financial accounting are using the same method/materials for recording and analyzing the transactions, the two systems different in their purpose and scope. Main points of differences are given below:

Basis	Financial Accounting	Cost Accounting
1. Objective	It provides information about the financial performance	It provides information of ascertainment of cost to

	and financial position of the business	control cost and for decision making about the cost.
2. Nature	It classifies records, presents and interprets transactions in terms of money.	It classifies, presents and interprets in a significant manner the material, labour and overheads cost
3. Recording of data	It records Historical data	It also records and presents the estimated/budgeted data. It makes use of both the historical costs and predetermined costs.
4. Users of information	The users of financial accounting statements are shareholders, creditors, financial analysts and government and its agencies etc.	The cost accounting information is used by internal management at different level.
5. Analysis of costs and profits	It shows the profit/loss of the organization	It provides the details of cost and profit of each product, process, job, contracts etc.
6. Time period	Financial statements are prepared for a definite period, usually a year.	Its reports and statements are prepared as and when required.
7. Presentation of information	A set format is used for presenting financial information	There are not any set formats for presenting cost information.

Distinguish between Financial Accounting and Management Accounting.

The Financial Accounting is the process of recording, summarizing and reporting the myriad of transactions from a business, so as to provide an accurate picture of its financial position and performance. The primary objective of financial accounting is the preparation of financial statements - including the balance sheet, income statement and cash flow statement - that summarizes the company's operating performance over a particular period, and financial position at a specific point in time. These statements - which are generally prepared quarterly and annually, and in accordance with Generally Accepted Accounting Principles (GAAP) - are aimed at external parties including investors, creditors, regulators and tax authorities.

Management accounting is a field of accounting that analyzes and provides cost information to the internal management for the purposes of planning, controlling and decision making.
Management accounting refers to accounting information developed for managers within an organization. CIMA (Chartered Institute of Management Accountants) defines Management accounting as "Management Accounting is the process of identification, measurement, accumulation, analysis, preparation, interpretation, and communication of information that used by management to plan, evaluate, and control within an entity and to assure appropriate use of an accountability for its resources". This is the phase of accounting concerned with providing information to managers for use in planning and controlling operations and in decision making.
Managerial accounting is concerned with providing information to managers i.e. people inside an organization who direct and control its operations. In contrast, financial accounting is concerned with providing information to stockholders, creditors, and others who are outside an organization. Managerial accounting provides the essential data with which organizations are actually run. Financial accounting provides the scorecard by which a company's past performance is judged.

Difference between Financial Accounting and Management Accounting:

Basis	Financial Accounting	Management Accounting
External vs. Internal	A financial accounting system produces information that is used by parties external to the organization, such as shareholders, bank and creditors.	A management accounting system produces information that is used within an organization, by managers and employees.
Segment reporting	Pertains to the entire organization or materially significant business units.	May pertain to smaller business units or individual departments, in addition to the entire organization.
Focus	Financial accounting focuses on history.	Management accounting focuses on future & present.
Format	Financial accounts are supposed to be in accordance with a specific format, so that financial accounts of different organizations can be easily compared. (Formal recordkeeping)	No specific format is designed for management accounting systems. (Formal and informal recordkeeping)

Planning and control	Financial accounting helps in making investment decisions, and in credit rating.	Management accounting helps management to record, plan and control activities to aid decision-making process.
Information	Quantitative and monetary	Quantitative and qualitative; Monetary and non-monetary
Users	Financial accounting reports are primarily used by external users, such as shareholders, bank and creditors.	Management accounting reports are exclusively used by internal users' viz. managers and employees.
Reporting frequency and duration	Well-defined - annually, semi-annually, quarterly. (Verifiable)	As needed - daily, weekly, monthly.
Optional?	Preparing financial accounting reports are mandatory especially for limited companies.	There are no legal requirements to prepare reports on management accounting.
Objectives	The main objectives of financial accounting are :i) to disclose the end results of the business, and ii) to depict the financial condition of the business on a particular date.	The main objectives of Management Accounting are to help management by providing information that used by management to plan, evaluate, and control.
Legal/rules	Drafted according to GAAP - General Accepted Accounting Procedure.	Drafted according to management suitability.
Accounting process	Follows a full process of recording, classifying, and summmarising for the purpose of analysis and interpretation of the finnancial information.	Cost accounts are not preserved under Management Accounting. The necessary data from financial statements and cost ledgers are analyzed.

Marginal costing as a technique of cost control.

The costs that vary with a decision should only be included in decision analysis. For many decisions that involve relatively small variations from existing practice and/or are for relatively limited periods of time, fixed costs are not relevant to the decision. This is because either fixed costs tend to be impossible to alter in the short term or managers are reluctant to alter them in the short term.

Marginal costing distinguishes between fixed costs and variable costs as conventionally classified. Marginal costing is an accounting technique. Marginal costing is not a system of costing such as process or job costing. As marginal costing is a technique, it may be used in conjunction with any costing method. Marginal costing as a technique helps the management to measure the profitability of an undertaking by considering the underlying cost behaviour. Marginal costing is an impotent technique, which guides the management theory. Marginal costing is also known as direct costing or variable costing or differential costing or incremental costing or comparative costing.

Definition:

According to ICMA London ³marginal cost is the amount for any given volume of output by which aggregate costs are changed if the volume of output is increased or decreased by one unit´. Marginal costing is the technique of applying the concept of marginal cost in decision making process. Marginal costing is a technique that distinguishes between fixed and variable costs. The ³marginal´ cost of a product is its variable cost.

How is the concept of marginal costing practically applied to control the cost increase the profit?

The concept of marginal costing is practically applied in the following situations:

1. Evaluation of Performance : The evaluation of the performance of various departments or products can be evaluated with the help of marginal costing which is based on contribution generating capacity.

2. Profit Planning: This technique through the calculation of P/V Ratio helps the management to plan the activities in such a way that the profit can be maximized.

3. Fixation of Selling Price: The technique of marginal costing assists the management to fix the price in such a way so that prices fixed can cover at least the variable cost.

4. **Make or Buy decision**: Marginal cost analysis helps the management in making or buying decision.

5. **Optimizing Product Mix**: To maximize profits and increase sales volume it is necessary to decide an optimized mix or proportion in which various products of a company can be sold.

6. **Cost Control**: Marginal Costing is a technique of cost classification and cost presentation which enable the management to concentrate on the controllable costs.

7. Flexible Budget preparation: As the marginal costing particularly classifies costs as fixed and variable costs which facilitates the preparation of flexible budgets.

Applications of Marginal Costing

To Check the Effect of Reducing of Current Price on profit

We all know, this is the time of competition, customer has become king. He wants product at minimum price. One example, we can see free video on YouTube. Instead of buying costly CDs and DVD, customers of entertainment industry see free films and movies on YouTube. But on the other side, company wants to maintain his current profit. At that time, manager will be in tension because it is not possible to maintain profit even after reducing price. But if manager learns marginal costing techniques and uses it effective way, they can check the effect of reducing of current price on net profit, after this, he can decide to reduce production or increase production. It is the law of economics, variable cost will reduce by reducing units of production in same proportion but when we increase production, fixed cost will fastly decreases due to constant nature.

To Choose of Good Product Mix

It may be possible that company is producing more than one product, at that time company has to calculate each product's contribution margin or gross profit margin. After this, manager see which product is giving high contribution margin. Company manager will give preference to that product whose contribution will high. One more decision can be taken by manger. He can check contribution by producing different quantity of different products. If he see any quantity of products is producing maximum contribution, it will be equilibrium point. Production of units at that quantity will be benefited to company

.

To Calculate of Margin of Safety

Marginal costing can be utilized for calculating margin of safety. Margin of safety is difference between actual sale and sale at breakeven point. According to marginal costing rules, production will follow sales. Suppose current sale is frs. 4, 00,000 and BEP is frs. 3, 00,000, margin of safety is frs.100000. We can calculate it with following formula;

= Profit/ P/V ratio

If company's sale is less than margin of safety, then manager can take step to reduce both fixed and variable cost or increase prices.

For Decision Regarding to sell goods at Different Prices to Different Customers

Sometime, company has to give special discount to special customers. These customers may be govt, foreign companies or wholesaler. At that time manager has to take decision at what limit, we can give discount to special customers. Marginal costing may help in this decision.

What is "Marginal Costing"? Discuss its advantages and disadvantages.

It is a costing technique based on the segregation of cost into fixed and variable. In this technique, only variable manufacturing costs are considered while determining the cost of goods sold and also for valuation of inventories. Fixed expenses are written off against the 'contribution' for that period. Contribution is the difference between the sales value and variable cost. If the company is producing more than one product, the contribution from each product is combined as a pool from which the total fixed cost is deducted to obtain the profit.

Definition:

According to ICMA London ³marginal cost is the amount for any given volume of output by which aggregate costs are changed if the volume of output is increased or decreased by one unit´. Marginal costing is the technique of applying the concept of marginal cost in decision making process. Marginal costing is a technique that distinguishes between fixed and variable costs. The marginal´ cost of a product is its variable cost.

Advantages and Disadvantages of Marginal Costing Technique

Advantages

1. Marginal costing is simple to understand.
2. By not charging fixed overhead to cost of production, the effect of varying charges per unit is avoided.
3. It prevents the illogical carry forward in stock valuation of some proportion of current year's fixed overhead.
4. The effects of alternative sales or production policies can be more readily available and assessed, and decisions taken would yield the maximum return to business.
5. It eliminates large balances left in overhead control accounts which indicate the difficulty of ascertaining an accurate overhead recovery rate.
6. Practical cost control is greatly facilitated. By avoiding arbitrary allocation of fixed overhead, efforts can be concentrated on maintaining a uniform and consistent marginal cost. It is useful to various levels of management.
7. It helps in short-term profit planning by breakeven and profitability analysis, both in terms of quantity and graphs. Comparative profitability and performance between two or more products and divisions can easily be assessed and brought to the notice of management for decision making.

Disadvantages

1. The separation of costs into fixed and variable is difficult and sometimes gives misleading results.

2. Normal costing systems also apply overhead under normal operating volume and this shows that no advantage is gained by marginal costing.
3. Under marginal costing, stocks and work in progress are understated. The exclusion of fixed costs from inventories affect profit, and true and fair view of financial affairs of an organization may not be clearly transparent.
4. Volume variance in standard costing also discloses the effect of fluctuating output on fixed overhead. Marginal cost data becomes unrealistic in case of highly fluctuating levels of production, e.g., in case of seasonal factories.
5. Application of fixed overhead depends on estimates and not on the actuals and as such there may be under or over absorption of the same.
6. Control affected by means of budgetary control is also accepted by many. In order to know the net profit, we should not be satisfied with contribution and hence, fixed overhead is also a valuable item. A system which ignores fixed costs is less effective since a major portion of fixed cost is not taken care of under marginal costing.
7. In practice, sales price, fixed cost and variable cost per unit may vary. Thus, the assumptions underlying the theory of marginal costing sometimes becomes unrealistic. For long term profit planning, absorption costing is the only answer.

Pass the following Journal entries in the Books of Madhura

June, 2012 1. Invested cash frs. 1,00,000 into the business.

1. Sold goods worth frs.12,00,000 at 5% trade discount to Radhika.
2. Borrowed frs.2, 00,000 from Amey.
3. Paid for stationery frs.15, 000.
4. Withdrawn cash for personal use frs.5, 000.
5. Purchased goods worth frs.8,00,000 from Ram & Co,
6. Received commission frs.18,000
7. Returned goods to Ram & Co. worth frs.20, 000.
8. Paid cash frs.78, 000 to Sudhir in full-settlement of frs. 80,000.

Answer:

Journal Entries

In the books of Madhura

Date	Particulars	L/F	Debit frs.	Credit frs.
June 2012				
1.	Cash A/C____Dr. To Capital A/C		1,00,000	 1,00,000

	(Being business started)			
2.	Radhika's A/C____Dr. To Sales A/C (Being goods sold on credit at 5% Trade Discount)		11,40,000	 11,40,000
3.	Cash A/C_____Dr. To Amey's Loan A/C (Being loan taken)		2,00,000	 2,00,000
4.	Stationery A/C__________Dr. To Cash A/C (Being stationery purchased)		15,000	 15,000
5.	Drawings A/C_______Dr. To Cash A/C (Being cash withdrawn for personal use)		5,000	 5,000
6.	Purchases A/C_______Dr. To Ram & Co. A/C (Being goods purchased on credit)		8,00,000	 8,00,000
7.	Cash A/C_______Dr. To Commission A/C (Being commission received)		18,000	 18,000
8.	Ram & Co. A/C_______Dr. To Purchase Returns A/C (Being purchased goods returned)		20,000	 20,000
9.	Sudhir's A/C_________Dr. To Cash A/C To Discount A/C (Being Sudhir paid in Full Settlement)		80,000	 78,000 2,000
			1,38,000	1,38,000

For production of 20,000 units the following are the budgeted expenses;

	Per Unit (frs.)
Direct Materials	500
Direct Labour	300
Variable Overheads	250
Fixed Overheads (frs.20,00,000)	100
Variable Expenses (Direct)	80

Selling Expenses (10% Fixed)	200
Administrative Expenses (frs.5,00,000 rigid for all levels of production)	25
Distribution Expenses (20% fixed)	150

Prepare a Flexible Budget for production of 22,000, 25,000 units showing distinctly marginal cost and total cost.

Answer:

Flexible Budget

Particulars	Per Unit (frs.)	Budget for 22,000 units	Budget for 25,000 units
Variable Costs			
Direct Materials	500	1,10,00,000	1,25,00,000
Direct Labour	300	66,00,000	75,00,000
Variable Overheads	250	55,00,000	62,50,000
Variable Expenses (Direct)	80	17,60,000	20,00,000
Selling Expenses (90% Variable)	200	39,60,000	45,00,000
Distribution Expenses (80% Variable)	150	26,40,000	30,00,000
Total Marginal Cost		**3,14,60,000**	**3,57,50,000**
Fixed Costs			
Fixed Overheads (frs.20,00,000)	100	20,00,000	20,00,000
Administrative Expenses (frs.5,00,000 rigid for all levels of production)	25	5,00,000	5,00,000
Selling Expenses (10% Fixed)	200	4,00,000	4,00,000
Distribution Expenses (20% fixed)	150	6,00,000	6,00,000
Total Fixed Cost		**35,00,000**	**35,00,000**
Total Costs			

You are given the following data:-

Year	Sales	Profit
2012	12, 00,000	80,000
2013	14, 00,000	1,30,000

Find out – i) P/V Ratio ii) B.E. Point iii) Profit when Sales are frs.18,00,000/-
iv) Sales are require to earn profit of frs. 1, 20,000/- v) Margin of Safety in 2012.

Answer :-

i. **P/V Ratio** = $\frac{\text{Change in Profit}}{\text{Change in Sales}} \times 100$

$= \frac{1,30,000-80,000}{14,00,000 - 12,00,000} \times 100$

$= \frac{50,000}{2,00,000} \times 100$

= 25%

ii. **Fixed Costs = (Sales X P/V Ratio) – Profit**

= (12,00,000 X 25%) – 80,000
= 3,00,000 – 80,000
= 2,20,000

BEP = $\frac{\text{Fixed Costs}}{\text{P/V Ratio}}$

$= \frac{2,20,000}{25\%}$

= 8,80,000

iii. **Profit when sales are frs.18,00,000/-**

Desired Profit = (Sales X P?V Ratio) – Fixed Costs
= (18,00,000 X 25%) – 2,20,000
= 4,50,000 – 2,20,000
= 2,30,000/-

iv. **Sales are require to earn profit of frs. 1, 20,000/-**

$$\text{Desired Sales} = \frac{\text{Fixed Cost + Profit}}{\text{P/V Ratio}}$$

$$= \frac{2,20,000 + 1,20,000}{25\%}$$

$$= \frac{3,40,000}{25\%}$$

= 13,60,000/-

v. **Margin of Safety in 2012.**

Margin of Safety = Sales – Sales at BEP
= 12,00,000 – 8,80,000
= 3,20,000

The following information is given:

Standard hours per unit	15 hours
Standard rate	frs.4 per hour
Actual production	1000 units
Actual hours	15300 hours
Actual rate	frs.3.90 per hour
Idle time	200 hours

Calculate : a) Labour Cost Variance b) Labour Rate Variance c) Labour Efficiency Variance d) Idle Time Variance

Answer :-

Standard For Actual			Actual		
Hours	Rate	Total		Rate	Total
15000	4	60000	15300	3.90	59670

					200

i. Labour Cost Variance = Std. Cost for Actual Output – Actual Costs
= 60000 – 59,670
= 330 (Favourable)

ii. Labour Rate Variance = (Std. Rate – Actual Rate) X Actual Hours
= (4 – 3.90) X 15300
= 1,530 (Favourable)

iii. Labour Efficiency Variance = (Std. Hours for Actual Output – Actual Hours) X Std. Rate
= (15000 – 15300) X 4
= 1,200 (Adverse)

Iv Idle Time Variance = Idle Hours X Std. Hourly Rate
= 200 X 4
= 800 (Adverse)

CHAPTER SEVEN

BUDGETARY CONTROL

What is "Budgetary Control"? Explain the objectives of Budgetary Control System.

Many people think of budgeting as something to do when they're short on cash. College students might turn to a budget to figure out how to make due with their high expenses and limited incomes. Wise new grads create budgets to make sure they're properly allocating their first paychecks among emergency savings, retirement savings, student loan repayments, rent and utilities, and rewards for their hard work like new gadgets and nights on the town. Young couples trying to figure out how to afford a wedding, or newlyweds wondering how to fit the expense of buying a house or having a child into their monthly cash flow, are also likely to make budgets. Of course, budgets are commonly associated with people of all ages who are barely able to make ends meet.

The truth is that budgeting isn't just for times when your money is tight or your life is undergoing a major transition. Budgeting is for everyone, rich and poor alike. In fact, budgeting will be that much easier in times of change if you do it all the time. Where do you think a Fortune 500 company like Amazon would be today without proper budgeting? What about wealthy people like Warren Buffett? There's no way that he or his holding company, Berkshire Hathaway, could have achieved such success without paying attention to their monthly, quarterly and annual cash inflow and outflow. Budgeting won't just get you out of a rut - it can also help you get rich.

Definition of 'Budget' and Budgeting:

A **budget** is a quantitative expression of a plan for a defined period of time. It may include planned sales volumes and revenues, resource quantities, costs and expenses, assets, liabilities and cash flows. It expresses strategic plans of business units, organizations, activities or events in measurable terms.

An estimation of the revenue and expenses over a specified future period of time. A budget can be made for a person, family, group of people, business, government, country, multinational organization or just about anything else that makes and spends money.

A budget is a microeconomic concept that shows the tradeoff made when one good is exchanged for another.

Budgeting for a business is a process. It is the process of preparing a detailed statement of financial results that are expected for a given time period in the future. There are two keywords in that statement. The first keyword is "expected." Expected means something that is likely to happen. The second keyword is "future" which is a period in the time to come. So, budgeting is the process of preparing a detailed statement of financial results that are likely to happen in a period in a time to come.

Budgetary Control System:

Methodical control of an organization's operations through establishment of standards and targets regarding income and expenditure, and a continuous monitoring and adjustment of performance against them.

Objectives of Budgetary Control System:

The following are the objectives of Budgetary Control System;

1. **Planning:** A budget provides a detailed plan of action for a business over a definite period of time. Detailed plans relating to production, sales, raw material requirements, labour needs, advertising and sales promotion performance, research and development activities, capital additions etc. are drawn up. By planning many problems are anticipated long before they arise and solutions can be sought through careful study. Thus most business emergencies can

be avoided by planning. In brief, budgeting forces the management to think ahead, to anticipate and prepare for the conditions.

2. **Co-ordination**: Budgeting aids managers in co-ordinating their efforts so that objectives of the organization as a whole harmonise with the objectives of the divisions. Effective planning and organization contributes a lot in achieving coordination. There should be co-ordination in the budgets of various departments. For example, the budget of sales should be in co-ordination the purchase budget. Similarly, production budget should be prepared in co-ordination with the purchase budget and so on.
3. **Communication**: A budget is a communication device. The approved budget copies are distributed to all management personnel which provides not only adequate understanding and knowledge of the programmes and policies to be followed but also gives knowledge about the restrictions to be adhered to. It is not the budget itself that facilitates communication, but the vital information is communicated in the act of preparing budgets and participation of all responsible individuals in this act.
4. **Motivation**: A budget is useful device for motivating managers to perform in line with the company objectives. If individuals have actively participated in the preparation of budgets, it acts as a strong motivating force to achieve the targets.
5. **Control**: Control is necessary to ensure that plans and objectives as laid down in the budgets are being achieved. Control, as applied to budgeting, systematized effort to keep the management informed of whether planned performance is being achieved or not. For this purpose, a comparison is made between plans and actual performance. The difference between the two is reported to the management for taking corrective action.
6. **Performance evaluation**: A budget provides a useful means of informing managers how well they are performing in meeting targets they have previously helped to set. In many companies, there is a practice of rewarding employees on the basis of their achieving the budget targets or promotion of a managers may be linked to his budget achievement record.

What is “Budgeting?” Discuss the advantages and disadvantages of Budgetary Control System.

A **budget** is a quantitative expression of a plan for a defined period of time. It may include planned sales volumes and revenues, resource quantities, costs and expenses, assets, liabilities and cash flows. It expresses strategic plans of business units, organizations, activities or events in measurable terms.

An estimation of the revenue and expenses over a specified future period of time. A budget can be made for a person, family, group of people, business, government, country, multinational organization or just about anything else that makes and spends money.

A budget is a microeconomic concept that shows the tradeoff made when one good is exchanged for another.

Budgeting for a business is a process. It is the process of preparing a detailed statement of financial results that are expected for a given time period in the future. There are two keywords in that statement. The first keyword is "expected." Expected means something that is likely to happen. The second keyword is "future" which is a period in the time to come. So, budgeting is the process of preparing a detailed statement of financial results that are likely to happen in a period in a time to come.

Advantages:

1. **Efficiency**: Effective budgetary control system enables management of business concern to conduct its business activities in very efficient manner.
2. **Control of Expenses**: It is powerful tool utilized by business houses for the control of their expenses. It in fact provides yardstick and evaluates the performance of individuals and their departments.
3. **Finding deviations:** It reveals deviations to management from the budgeted figures after making a comparison with actual figures.
4. **Effective utilization of resources:** Effective utilization of various resources like men, materials, machinery, money, matters and market- is made possible, as the production is budgeted after taking them into consideration.
5. **Revision of Plans:** It helps in the review of current trends and framing of future policies.
6. **Implementation of Standard Costing System:** It creates suitable condition for the implementation of in the business organization.
7. **Cost Consciousness**: Budgets are studied by outside fund providers also such as banking and financial institutions, realizing that management encourages the cost consciousness and maximum utilization of available resources.
8. **Credit Rating:** Management which have deployed well-ordered budget plans and which operates accordingly, receives greater credit from credit rating agencies.

Limitations of Budgetary Control

The list of advantages given in the previous chapter is impressive, but a budget is not a cure all for organizational ills. Budgetary control system suffers from certain limitations and those using the system should be fully aware of them.

1. The budget plan is based on estimates. Budgets are based on forecasts and forecasting cannot be an exact science. Absolute accuracy, therefore, is riot possible in forecasting and budgeting. The strength -or weakness of the budgetary control system depends to a large extent, on the accuracy with which estimates are made. Thus, while using the system, the fact that budget is based on estimates must be kept in view.

2. Danger of rigidity. A budget programme should be dynamic and continuously deal with the changing business circumstances. Budgets will lose most of their usefulness if they obtain rigidity and are not revised with the altering circumstances.

3. Budgeting is only an instrument of management. Budgeting cannot take the place of management however only a device of management is. The budget needs to be regarded not like a master, but as a servant. Occasionally it is believed that launch of a budget programme alone is actually sufficient to ensure its good results. Execution of a budget will not take place automatically. It is required that the entire organizational should participate enthusiastically in the programme for that realization of the budgetary objectives.

4. Expensive technique. The operation and installation of a budgetary manage system is a costly affair as it needs the employment of specialist staff and involves additional expenditure which small concerns might find hard to incur. However, it is vital that the cost of introducing an operating a budgetary control system must not exceed the rewards derived therefrom.

State how the "Standard Costing" is useful technique to control the costs.

An estimated or predetermined cost of performing an operation or producing a good or service, under normal conditions.

Standard costs are used as target costs (or basis for comparison with the actual costs), and are developed from historical data analysis or from time and motion studies. They almost always vary from actual costs, because every situation has its share of unpredictable factors. Also called normal cost.

Concept and Meaning of Standard Costing

Historical costs are the actual costs that are ascertained after they are incurred. During the initial stages of development of cost accounting, historical costing systems like process costing, contract costing, service costing etc. were available for ascertaining the costs. The historical costing methods are used to determine the cost incurred for the production of a particular products or completion of a particular job.

The historical costing systems suffer from several limitations; some of them are as follows:
* No basis for cost control
* No yardstick for measuring efficiency
* Delay in availability of information
* Expensive

Although standard costing attempts to overcome the limitations of historical costing system, it is not an alternative to the existing historical costing. Standard costing is the most widely used technique of controlling costs.

Standard costing is a technique that establishes predetermined estimates of the cost of products and services and compares these costs with the actual costs as they incur. Standard costing can be considered as a yardstick to measure the efficiency with the actual cost incurred. Hence, standard costing is system of costing which makes a comparison between the standard cost of each product or service with its actual cost to determine the efficiency of the operation, with a view to take corrective actions at the earliest possible time.

Thus, standard costing is the preparation and uses of standard costs, which involves the following process:
* Establishment of standard costs
* Ascertainment of actual costs
* Comparison of the above two and measurement of variances
* Analysis of variances
* Reporting to responsibility centers to take appropriate and necessary remedial actions.

Applications of Standard Costing

Standard costing is a useful method of control in a number of ways.

First, the process of evaluating performance by determining how efficiently current operations are being carried out may be facilitated by the process of management by exception. Very often the problem facing management is the time lost in sifting large masses of feedback information and in deciding what information is significant and relevant to the control problem. Management by exception overcomes this problem by highlighting only the important control information that is the variances between the standard set and the actual result. This process allows management to focus attention on important problems so that maximum energy may be devoted to correcting situations which are falling out of control.

Second, a standard costing system may lead to cost reductions. The installation of such a system demands a re-appraisal of current production methods as it necessitates the standardization of practices. This examination often leads to an improvement in the methods employed which is reflected in a reduction of the costs of the product. One example of cost reductions through increased efficiency may be seen in the simplication of the clerical procedures relating to inventory control. All similar items of inventory may be recorded in the accounts at a uniform price; this eliminates the need which arises under historical costing for re-calculating a new unit price whenever a purchase of inventory is made at a different price.

Third, standard costs are used as a basis for determining selling prices. Standard costs represent what the product should cost, and are a much better guide for pricing decisions than historical costs which may contain purchasing and production inefficiencies which cannot be recouped in competitive markets.

Finally, perhaps the most important benefit which may be derived from a standard costing system is the atmosphere of cost consciousness which is fostered among executives and foremen. Each individual is aware that the costs and output for which he is responsible are being measured, and that he will be called on to take whatever action is necessary should large variances occur. As we concluded earlier, if the philosophy of top management is positive and supportive, standard costing may act as an incentive to individuals to act in the best interest of the firm. Moreover, a standard costing system which allows subordinates to participate in setting the standards fosters a knowledge of costing down to shop floor level, and assists in decision making at all levels. Thus, if there should occur spoilt work necessitating a decision from the foreman in charge on whether to scrap or rectify the part involved, a knowledge of costs will enable him to make the best decision.

Define "Standard Costing". Discuss its advantages and disadvantages.

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Definition:
Standard costing is the practice of substituting an expected **cost** for an actual **cost** in the accounting records, and then periodically recording variances that are the difference between the expected and actual **costs**.

A management tool used to estimate the overall cost of production, assuming normal operations.

Advantages of Standard Costing
The following are the advantages of the standard costing system:

1. Management by Exception the standard costing is an example of management by exception. By studying the variances, management's attention is directed towards those items, which are not proceeding according to the plan. Most of the management's time is saved and can be directed to other value adding activities. Management only concentrates on the 'few' exceptions reported. **2. Cost Reduction** The process of setting, revising and monitoring standards encourages reappraisal of methods, materials and techniques thus leading to cost reductions. Analysis of unfavourable variances directs cost analysis to factors that are making costs to exceed the budgeted costs thus these factors can be controlled, leading to cost reduction.
3. Reliable base of calculating total Pricing Standard costs serve as a reliable base of calculating total cost of producing a good or service, to which a margin can be added to determine the selling price.
4. Inventory Valuation Standard costing makes inventory valuation much easier, if the actual number of physical units in the inventory is known, then the inventory value is simply determined by multiplying the standard cot per unit by the physical units.
5. Motivation A properly developed standard costing system requires the full participation of all management levels (upper, middle and lower levels) and the employees. This creates motivation for the employees as they feel part of the system.
6. Cost Control A well implemented standard costing system acts as a yardstick against which all costs are measured to determine whether the variance from the standard is favourable or unfavourable. This creates cost consciousness in the organization and in the end enables the organization to control costs.
7. Budgeting is made easier One of the greatest benefits of standard costing is to be found in setting budgets for the organization and its departments. As earlier illustrated once the desired

output units are known, then the budgeted cost is simply the output units desired multiplied by the standard cost per unit.

Disadvantages of Standard Costing

1. Expensive: The system of standard costing is expensive to install A lot of money is spent in studying output requirements in terms of labour, materials and overheads.

2. Time Consuming A lot of time is also spent in developing and installing reliable standard costing systems.

3. Obsolescence: In fast changing conditions (e.g. in hyperinflationary economies where prices of labour, materials and overheads change rapidly), standards become out of date quickly. They therefore lose their control and motivational effects.

4. Hard to Understand Some standard costing systems are overly elaborate and are therefore not well understood by line managers and employees. This makes their implementation difficult.

5. Effectiveness depends on Environment For standard costing systems to be effective in cost control and performance evaluation, then a participative and democratic management style is required. The top management and employees need to be committed to attaining the set standards of performance. An effective and efficient management information system is also required so as to provide employees and managers with reliable, accurate and timely feedback regarding their performance. Lack of one or more of these requirements frustrates the success of a standard costing system so that its effectiveness cannot be realized.

6. It is Subjective As we have already seen, there are several types of standards that an organization can adopt (basic, ideal, attainable and cement). What is therefore a standard in an organization depends on its management. It is also important to note that what is referred to as a significant variance depends on the organization's management, thus the subjectivity. If this subjectivity is poorly managed, (for example, punishing employees for insignificant unfavourable variances of for variances arising from factors beyond their control), then a standard costing system can lead to employee frustration and poor goal congruence in the organization.

Discuss the various types of Budgets in detail.

A **budget** is a quantitative expression of a plan for a defined period of time. It may include planned sales volumes and revenues, resource quantities, costs and expenses, assets, liabilities and cash flows. It expresses strategic plans of business units, organizations, activities or events in measurable terms.

An estimation of the revenue and expenses over a specified future period of time. A budget can be made for a person, family, group of people, business, government, country, multinational organization or just about anything else that makes and spends money.

A budget is a microeconomic concept that shows the tradeoff made when one good is exchanged for another.

Budgeting for a business is a process. It is the process of preparing a detailed statement of financial results that are expected for a given time period in the future. There are two keywords in that statement. The first keyword is "expected." Expected means something that is likely to happen. The second keyword is "future" which is a period in the time to come. So, budgeting is the process of preparing a detailed statement of financial results that are likely to happen in a period in a time to come.

Types of Functional Budgets

Master Budget

A master budget is a comprehensive projection of how management expects to conduct all aspects of business over the budget period, usually a fiscal year. The master budget summarizes projected activity by way of a cash budget, budgeted income statement and budgeted balance sheet. Most master budgets include interrelated budgets from the various departments. Managers typically use these subset budgets to plan and set performance objectives. Master budgets are generally used in larger businesses to keep many managers on the same page.

Capital Budgets

A capital budget estimates all capital asset acquisitions and summarizes all expenses and costs of major purchases for the next year. Capital assets include items that have useful lives of more than 12 months, such as buildings, building improvements, land, furniture, fixtures, equipment, computers, musical instruments, works of art and books, writes David C. Maddox, the author of the book "Budgeting for Not-for-Profit Organizations." The main purpose of a capital budget is to forecast costs of major capital purchases.

Operating Budgets

Operating budgets indicate the products and services a firm expects to use in a budget period. It describes all the income-generating activities of a firm, including production, sales and inventories of finished goods. An operating budget typically has two distinct parts: the expense budget and the revenue budget. The expense budget indicates all expected expenses of a firm for the coming year, while the revenue budget shows all projected revenues for the coming year.

Cash Budgets

A cash budget projects all cash inflows and outflows for the next year. Cash budgets have four distinct elements: cash disbursements, cash receipts, net change in cash and new financing, writes Arthur J. Keown in the book “Foundations of Finance.” A cash budget is important, because it allows administrators to timely identify periods with cash overages and shortages so they can take necessary remedial action.

Sales Budgets

Sales budgets indicate the sales a firm expects to make in units and dollars for a budget year. They detail the quantities of products or services a firm expect to sell, revenues incurred from those sales and all expenses accrued during selling. A sales budget is a planning instrument and a control mechanism. Sales budget forecasts determine sales potential, or the maximum number of sales a firm can make. This information is then used to plan resource allocations to achieve those sales levels. Sales budgets serve as benchmarks or yardsticks against which actual sales performance is measured and variables such as sales volume, profitability and selling expenses are controlled.

Personnel Budgets

Personnel budgets, or salary and wage budgets, are cost estimations related to labor. They forecast the costs of recruitment, hiring, training, assignment, salaries, overtime costs, additional benefits and discharge. Calculating personnel budgets includes estimating the number of staff, staffing ratios and overheads, write George M. Guess and Paul G. Farnham, the authors of the book, “Cases in Public Policy Analysis.”

What is a Cost Sheet? Give an example of cost sheet clearly indicating the headings and the important items supplying imaginary figures.

A **cost sheet** is a report on which is accumulated all of the **costs** associated with a product or production job. A **cost sheet** is used to compile the margin earned on a product or job, and can form the basis for the setting of prices on similar products in the future.

A document that reflects the cost of the items and services required by a particular project or department for the performance of its business purposes. For example, a departmental cost sheet might include the material costs, labor costs and overhead costs incurred over a given time frame by a department and it therefore provides a record of costs that are chargeable to that department.

Cost Sheet is a detailed statement of the elements of cost incurred in production, arranged in a logical order under different heads such as materials, labor and overheads, prepared at short intervals of time.

COST SHEET OF ZYX LTD.

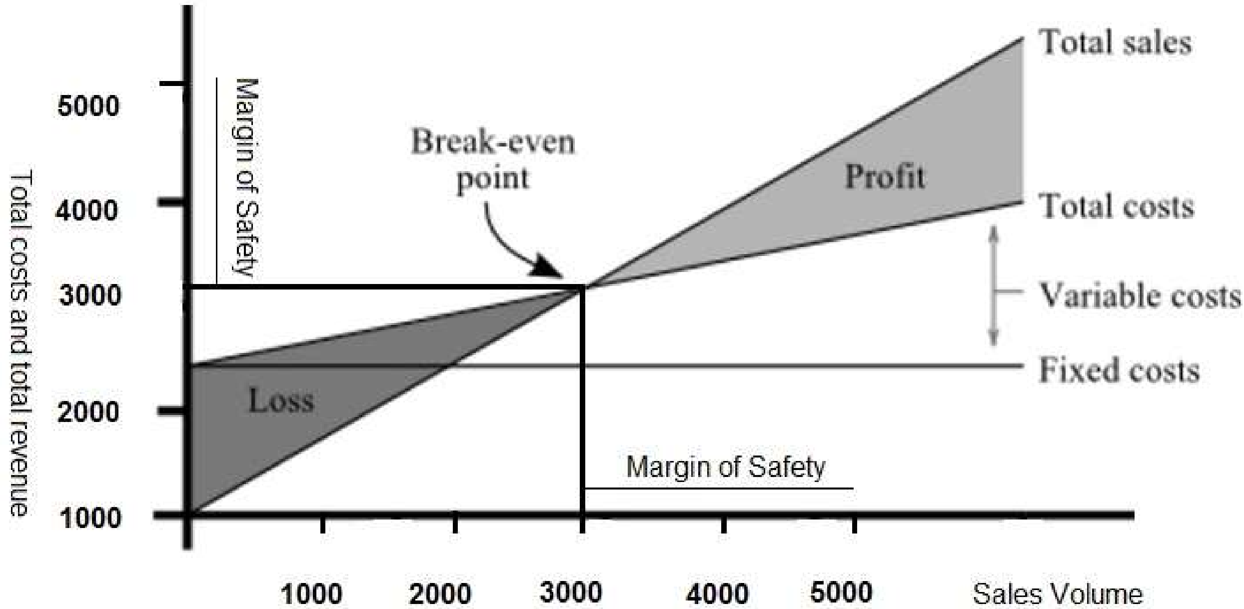
For the financial year 2013-14

Particulars	Amount	Amount
Opening Stock of Raw Material	188000	
Add: Purchase of Raw materials	832000	
Add: Purchase Expenses	2000	
	1022000	
Less: Closing stock of Raw Materials	200000	
Raw Materials Consumed	822000	
Direct Wages (Labour)	238000	
Direct Charges	50000	
Prime cost (1)		1110000
Add :- Factory Over Heads:		
Factory Rent	10000	
Factory Power	15000	
Indirect Material	5000	
Indirect Wages	35000	
Supervisor Salary	20000	
Factory Insurance	1500	
Factory Asset Depreciation	15000	101500
Works cost Incurred		1211500
Add: Opening Stock of WIP	120000	
Less: Closing Stock of WIP	95000	25000
Works cost (2)		1236500
Add:- Administration Over Heads:-		
Office Rent	12000	
Asset Depreciation	15000	
General Charges	8000	
Audit Fees	12000	
Bank Charges	900	
Other Office Expenses	2000	49900
Cost of Production (3)		1286400
Add: Opening stock of Finished Goods	201000	
Less: Closing stock of Finished Goods	165000	36000
Cost of Goods Sold		1322400
Add:- Selling and Distribution OH:-		
Salesmen's Commission	22000	

Salesmen's salary Traveling Expenses Advertisement Deliverymen's expenses Sales Tax	15000 7000 20000 5000 8500	77500
Cost of Sales (5)		1399900
Profit (balancing figure)		100100
Sales		1500000

What is Break-Even Analysis? Write assumptions and uses of Break-Even Analysis.

Break-even analysis is a supply-side analysis; that is, it only analyzes the costs of the sales. It does not analyze how demand may be affected at different price levels. For example, if it costs $50 to produce a widget, and there are fixed costs of $1,000, the break-even point for selling the widgets would be:
If selling for $100: 20 Widgets (Calculated as 1000/(100-50)=20)
If selling for $200: 7 Widgets (Calculated as 1000/(200-50)=6.7)
An analysis to determine the point at which revenue received equals the costs associated with receiving the revenue. Break-even analysis calculates what is known as a margin of safety, the amount that revenues exceed the break-even point. This is the amount that revenues can fall while still staying above the break-even point.
Break-even analysis is a technique widely used by production management and management accountants. It is based on categorizing production costs between those which are "variable" (costs that change when the production output changes) and those that are "fixed" (costs not directly related to the volume of production). Total variable and fixed costs are compared with sales revenue in order to determine the **level of sales volume, sales value or production at which the business makes neither a profit nor a loss (the "break-even point").**
The Break-Even Chart
In its simplest form, the break-even chart is a graphical representation of costs at various levels of activity shown on the same chart as the variation of income (or sales, revenue) with the same variation in activity. The point at which neither profit nor loss is made is known as the "break-even point" and is represented on the chart below by the intersection of the two lines:



In the diagram above, the line OA represents the variation of income at varying levels of production activity ("output"). OB represents the total fixed costs in the business. As output increases, variable costs are incurred, meaning that total costs (fixed + variable) also increase. At low levels of output, Costs are greater than Income. At the point of intersection, P, costs are exactly equal to income, and hence neither profit nor loss is made.

Assumption of Break-Even Analysis:

1. It assumes that selling prices are constant no matter the level of output.
2. It assumes that all productions are sold without opening/closinginventories.
3. It assumes that fixed costs remain fixed at all levels of output.
4. It assumes that variable costs per unit remain the same no matter theoutput level.
5. It can only apply to single product or a single mix of products.

Uses of Break-Even Analysis:

1. Measure profits and losses at different levels of activity.
2. Predict the effect of changes in the prices of the products.
3. Analyze the relationship between fixed costs and variable costs.
4. Predict the effect on profitability of changes in cost and efficiency.

CHAPTER EIGHT

VARIANCE

Discuss in detail the various types of variances.

Standard Costing guides as a measuring rod to the management for determination of "Variances" in order to evaluate the production performance. The term "Variances" may be defined as the difference between Standard Cost and actual cost for each element of cost incurred during a particular period. The term "Variance Analysis" may be defined as the process of analyzing variance by subdividing the total variance in such a way that management can assign responsibility for off-Standard Performance.

The variance may be favourable variance or unfavourable variance. When the actual performance is better than the Standard, it resents "Favourable Variance." Similarly, where actual performance is below the standard it is called as "Unfavourable Variance."

Variance analysis helps to fix the responsibility so that management can ascertain -

(a) The amount of the variance

(b) The reasons for the difference between the actual performance and budgeted performance

(c) The person responsible for poor performance

(d) Remedial actions to be taken

Variances are basically classified into following three categories:-

I. Direct Material Variance (DMV)

II. Direct Labour Variance (DLV)

III. Overhead Variance (OV)

I. Direct Material Variances

Direct Material Variances are also termed as Material Cost Variances. The Material Cost Variance is the difference between the Standard cost of materials for the Actual Output and the Actual Cost of materials used for producing actual output. If the actual costs is more than standard cost the variance will be unfavourable or adverse variance and. on tile other hand. if the actual cost is less than standard cost the variance will be favourabie variance. The material cost variance is further classified into: (I) Material Price Variance (2) Material Usage Variance (3) Material Mix Variance (4) Material Yield Variance

(1) Material Price Variance (MPV) : Material Price Variance is that portion of the Material Cost Variance which is due to the difference between the Standard Price specified and the Actual Price paid for purchase of materials. Material Price Variance may be calculated by

Material Price Variance = (Standard Price- Actual Price) x Actual Quantity

Note: If actual cost of materials used is more than the standard cost the variance is adverse. it represents negative (-) symbol. And on the other hand. if the variance is favourable it is to be represented by positive (+) symbol.

(2) Material Usage Variance (MUV): Material Usage Variance is that part of Material Cost Variance which refers to the difference between the standard cost of standard quantity of material for actual output and the Standard cost of the actual material used. Material Usage Variance is calculated as follows:

Material Usage Variance = (Std. Qty. for Actual Output – Actual Qty.) x Std. Price

Note: This Variance will be favourable when standard cost of actual material is more than the Standard material cost for actual output, and Vice Versa.

(3) Material Mix Variance (MMV) : It is the portion of the material usage variance which is due to the difference between the Standard and the actual composition of mix. Material Mix Variance is calculated under two situations as follows :

(a) When actual weight of mix is equal to standard weight to mix

(b) When actual weight of mix is different from the standard mix

(a) When Actual Weight and Standard Weight of Mix are equal.

(i) The formula is used to calculate the Variance:

Material Mix Variance = (Revised Std. Qty. – Actual Qty) x Std. Price

(4) Materials Yield Variance (MYV): It is the portion of Material Usage Variance. This variance arises due to spoilage, low quality of materials and defective production planning etc. Materials Yield Variance may be defined as "the difference between the Standard Yield Specified and the Actual Yield Obtained." This variance may be calculated as under:

Material Yield Variance =

(Actual Yield – Std. Yield from actual input) x Std. Material cost per unit

The following equations may be used for verification of Material Cost Variances :

(1) Material Cost Variance

(2) Material Usage Variance

(3) Material Cost Variance

= Material Price Variance + Material Usage Variance = Material Mix Variance - Material Yield Variance = Material Mix Variance + Material Yield Variance

II. Labour Variances

Labour Variances can be classified into:
(a) Labour Cost Variance (LCV)
(b) Labour Rate Variance or Wage Rate Variance
(c) Labour Efficiency Variance
(d) Labour Idle Time Variance
(e) Labour Mix Variance (0 Labour Revised Efficiency Variance
(g) Labour Yield Variance

(a) Labour Cost Variance (LCV): Labour Cost Variance is the difference between the Standard Cost of labour allowed for the actual output achieved and the actual wages paid. It is also termed as Direct Wage Variance or Wage Variance. Labour Cost Variance is calculated as follows: Labour Cost Variance = Standard Cost of Labour - Actual Cost of Labour
Note: If actual labour cost is more than the standard labour cost, the variance represents negative and vice versa.

(b) Labour Rate Variance: It is that part of labour cost variance which is due to the difference between the standard rate specified and the actual rate paid. This variances arise from the following reasons:
(a) Change in wage rate.
(b) Faulty recruitment.
(c) Payment of overtime.
(d) Employment of casual workers etc.
It is expressed as follows :
Labour Rate Variance = (Standard Rate- Actual Rate) x Actual Hours
Note: If the Standard rate is higher than the actual rate, the variance will be favourable and vice versa.

(c) Labour Efficiency Variance: Labour Efficiency Variance otherwise known as Labour Time Variance. It is that portion of the Labour Cost Variance which arises due to the difference between standard labour hours specified and the actual labour hours spent. The usual reasons for this variance are (a) poor supervision (b) poor working condition (c) increase in labour turnover (d) defective materials. It may be calculated as following:
Note: If actual time taken is more than the specified standard time, the variance represents unfavourable and vice versa.

(d) Labour Idle Time Variance: Labour Idle Time Variance arises due to abnormal situations like strikes, lockout, breakdown of machinery etc. In other words, idle time occurs due to the

difference between the time for which workers are paid and that which they actually expend upon production. It is calculated as follows :
Idle Time Variance = Idle Hours x Standard Rate

(e) Labour Mix Variance: It is otherwise known as Gang Composition Variance. This variance arises due to the differences between the actual gang composition than the standard gang composition. Labour Mix Variance is calculated in the same way of Materials Mix Variance. This variance is calculated in two ways:
(i) When Standard Labour Mix is equal to Actual Labour Mix.
(ii) When Standard Labour mix is different from Actual Labour Mix.
(i) When Standard and actual times of the labour mix are same The formula for its computation may be as follows :
Labour Mix Variance = { Standard Cost of Standard Labour Mix Standard Cost of } Actual Labour Mix
(ii) When Standard and actual times of the labour mix are different : Changes in the composition of a gang may arise due to shortage of a particular grade of labour. It may be calculated as follows:
Labour Mix Variance ={Revised Standard Time -Actual Time } x Standard Rate

(I) Labour Yield Variance: 'This variance is calculated in the same way as Material Yield Variance. Labour Yield Variance arises due to the variation in labour cost on account of increase or decrease in yield or output as compared to relative standard. The formula for this purpose is as follows:
Labour Yield Variance = (Actual Yield – Std. Yield from actual input) x Std. Labour cost per unit

Note: If actual output is more than Standard output for actual time, the variance is favourable and vice versa.
Verification: Labour Cost Variance = Labour Rate Variance + Labour Efficiency Variance

III. Overhead Variances
Overhead may be defined as the aggregate of indirect material cost, indirect labour cost and indirect expenses. Overhead Variances may arise due to the difference between standard cost of overhead for actual production and the actual overhead cost incurred.
Overhead Variances can be classified as :

a) Variable Overheads Variances

i) V. Overhead Cost Variance = (Std.Hrs.for Actual Output x Std.Overhead Rate) – Actual Overheads
ii) V. Overhead Expenditure Variance = (Actual Hrs. X Std. Rate) – Actual Overheads.
iii) V. Overhead Efficiency Variance =(Std. hrs.for actual output – Actual hrs)

X Std. Overhead Rate.

b) Fixed Overheads Variances

i) F. Overhead Cost Variance = (Std.Hrs.for Actual Output x Std.F.O.Rate) –
Actual Fixed Overheads

ii) F. Overhead Expenditure Variance = Budgeted F.O.- Actual F.Overheads.

iii) F. Overhead Volume Variance =(Std. hrs.for actual output – Budgeted hrs)
x Std. overhead Rate.